INCH POUND

MIL-DTL-83406E (USAF)
15 August 2024

SUPERSEDING
MIL-DTL-0083406D
15 April 2010

DETAIL SPECIFICATION

ANTI-G GARMENT, CUTAWAY, CSU-13B/P

Reactivated after 24 March 2006 and may be used for new and existing designs and acquisitions.

This specification is for an Aviation Critical Safety Item (CSI/Flight Safety Critical Aircraft Part (FSCAP) and the acquisition process must comply with the DoD Materiel Management Regulation – DoD 4140.1-R



Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, Bldg 28, 2145 Monahan Way, Wright-Patterson AFB OH 45433-7017 or emailed to Engineering_Standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1. Scope.

This specification covers the detail requirements and inspections for the CSU-13B/P anti-g garment, which is designated as a critical safety item.

1.2. CLASSIFICATION.

Anti-g garments are of the following classes and sizes, as specified (see 6.2 and 6.6):

Class 1 – Sage Green 1659
Class 2 – Tan 1657

Small regular (SR)	Medium regular (MR)	Large regular (LR)	Large extra-long (LXL)
Small long (SL)	Medium long (ML)	Large long (LL)	

2. APPLICABLE DOCUMENTS

2.1. General.

The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government documents.

2.2.1. Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50195	Thread, Aramid
A-A-55126	Fastener Tapes, Hook and Loop, Synthetic
A-A-55621	Fasteners, Snap, Style 3-Pronged Ring Head Type (16 Ligne)
A-A-55634	Zippers (Fasteners, Slide Interlocking)
A-A-55809	Insulation Tape, Electrical, 600V, Polyvinyl Chloride, Pressure-Sensitive Adhesive
A-A-59826	Thread, Nylon

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-83242	Cord, Aromatic Polyamide, Nonmelting
MIL-C-83390	Connector, Hose, Quick-Acting, Male, Anti-G Garment
MIL-C-83489	Cloth, Coated, Nylon, Polyurethane Coated
MIL-DTL-4088	Webbing, Textile, Woven Nylon (DoD Adopted)
MIL-DTL-5038	Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-DTL-10884	Fasteners, Snap
MIL-DTL-32075	Label: for Clothing, Equipage, and Tentage, (General Use)
MIL-DTL-32092	Leather, Cattlehide, Deerskin and Horsehide, Chrome Tanned
MIL-DTL-43204D	Cloth, Spacer (Olefin)
MIL-L-43283	Leather, Calfskin, Kip, and Cattlehide for Footwear Uppers, Chrome Tanned
MIL-PRF-6855/4	Rubber, Synthetic, Tubing
MIL-W-5664	Webbing, Textile, Elastic
MIL-W-81116	Webbing, Textile, Polyamide, High Temperature Resistant, Loop

DEPARTMENT OF DEFENSE STANDARDS

MS 27755	Connector, Hose Quick Disconnect, Female, Anti “G” Suit

(Copies of these documents are available online at https://assist.dla.mil or Defense Logistics Agency Troop Support, Clothing and Textiles Directorate online request form at https://www.dla.mil/Troop-Support.

2.2.2. Other Government documents, drawings, and publications.
The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE DRAWINGS

71380	Patterns, CSU-13B/P anti-g garment, Size Small Regular
71381	Patterns, CSU-13B/P anti-g garment, Size Small Long
71382	Patterns, CSU-13B/P anti-g garment, Size Medium Regular
71383	Patterns, CSU-13B/P anti-g garment, Size Medium Long
71384	Patterns, CSU-13B/P anti-g garment, Size Large Regular
71385	Patterns, CSU-13B/P anti-g garment, Size Large Long

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71386	Patterns, CSU-13B/P anti-g garment, Miscellaneous All Sizes
74204	Spacer, Tube – Anti-g Garment
74206	Pattern, Anti-g Garment, CSU-13B/P, Size Large Extra Long
200518653	Source Control Drawing- Fastener, Slide
201221034	Cloth, Plain Weave, Aramid

(Copies of these documents are available from the Defense Logistics Agency Troop Support, Clothing and Textiles Directorate online request form at https://www.dla.mil/Troop-Support or for DoD personnel they are available online at www.jedmics.net.)

CODE OF FEDERAL REGULATION

14 CFR Part 25 Aeronautics and Space

Defense Federal Acquisition Regulation Supplement, Clause 252.211-7003

(Copies of these documents are available online http://www.gpo.gov or from the Government Printing Office, 732 North Capitol Street NW, Washington, DC 20401-0001.)

2.3. Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NATIONAL AEROSPACE STANDARDS (NAS)

NASM20230	Grommet, Metallic, Plain and Rolled Rim, with Washer, Type I and Type III (DoD Adopted)
NASM27980	Fasteners, Snap, Style 2 (Regular Wire Spring Clamp Type) (DoD Adopted)
NASM27983	Fasteners, Snap, Style 4 (Three Way Locking Type) (DoD Adopted)

(Copies of these documents are available from www.aia-aerospace.org or Aerospace Industries Association, 1000 Wilson Blvd, Suite 1700, Arlington VA 22209-3901.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted)

(Copies of this document are available from www.asq.org or the American Society for Quality, 600 North Plankinton Ave., Milwaukee WI 53203-2914.)

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ASTM INTERNATIONAL

ASTM A313/A313M	Standard Specification for Stainless Steel Spring Wire (DoD Adopted)
ASTM D3990	Standard Terminology Relating to Fabric Defects
ASTM D5034	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D6193	Standard Practice for Stitches and Seams (DoD Adopted)

(Copies of these documents are available from www.astm.org or ASTM International, 100 BarR Harbor Dr., West Conshohocken PA 19428-2959.)

THE PARACHUTE INDUSTRY ASSOCIATION (PIA)

PIA-C-3953	Cloth, Duck, Nylon (DoD Adopted)
PIA-T-38328	Tape, Textile, Nylon, Aromatic, Nonmelting, Reinforcing

(Copies of these documents are available from www.pia.com or the Parachute Industry Association, 3833 West Oakton St., Skokie IL 60076.)

SAE INTERNATIONAL

SAE AMS-T-22085	Tapes, Pressure-Sensitive, Adhesive, Preservation and Sealing (DoD Adopted)

(Copies of this document are available from or www.sae.org or SAE World Headquarters, 400 Commonwealth Dr., Warrendale PA 15096-0001.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Test Method 135	Dimensional Changes of Fabrics after Home Laundering

(Copies of this document are available from or www.aatcc.org or AATCC, P.O. Box 12215 Research Triangle Park, NC 27709-2215)

2.4. Order of precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1. Qualification.

The anti-g garments furnished under this specification shall be products that are authorized by the qualifying activity for listing in the Qualified Products Database (QPD) before contract award (see 4.3 and 6.3).

3.2. First article.

When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

3.3. Recycled, recovered, or environmentally preferable materials.

Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. However, used, rebuilt, or refurbished items shall not be provided.

3.4. Materials and components.

Except for the metallic materials and components, the materials and components of the anti-g garment shall have been manufactured not more than 24 months prior to the date of acceptance of the anti-g garment by the Government. All materials and components shall be stored as specified by the material/component supplier and shall not be subjected to moisture, ultraviolet light, or environmental contaminants. Partially and fully assembled anti-g garments shall be stored to protect against damage from moisture, ultraviolet light, and environmental contaminants.

3.4.1. Basic materials

3.4.1.1. High temperature resistant cloth.

Except for the bladder, the basic material for the anti-g garment, the pull tape for the slide fasteners, the loops of the slide fasteners, and the hanger shall be plain weave conforming to 201221034 Type II, Class 1, and the color shall be AF Sage Green 1659 for Class 1 material and AF Tan 1657 for Class 2 material (see 6.4).

3.4.1.2. Polyurethane coated nylon cloth.

The material for the bladder, including the attachment patches, reinforcement of inflation tubes, external covering shall be plain weave conforming to MIL-C-83489 Type I, with the exception that finished fabric weight shall be 5.5 - 6.75 ounces per square yard and minimum tear strength shall be 5 lbs in the warp direction and 4 lbs in the fill direction.

3.4.2. Reinforcements and bindings.

Unless otherwise specified herein, the reinforcements shall be made of the basic material specified in 3.4.1.1. The centerfold, bias binding shall be 45 degrees bias, cut 1.25 inches to 1.75 inches wide from the basic material (see 3.4.1.1). Fold binding evenly in half to make a crease to mark the center point of the material. Unfold the binding and fold the two edges in to meet the crease made before. Fold the material back in half on the original crease to use for binding. See the binding in FIGURE 1A below. This operation can be completed with an automated machine with folder given that the binding is sewn to the suit catching all four layers of binding material and all layers of the base fabric.

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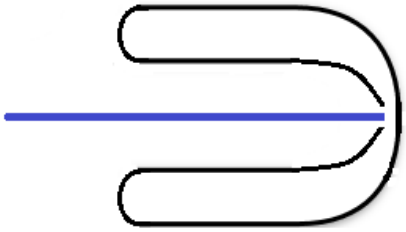


FIGURE 1A. Example of referenced binding.

3.4.3. Adjustment laces and lanyard.

The cord for the lanyard and for the adjustment laces shall be coreless conforming to MIL-C-83242 Type I and shall approximately match the shade of the basic material (see 3.4.1.1).

3.4.4. Slide fasteners.

The slide fasteners shall conform to TABLE I of this specification and AF Drawing 200518653, except as otherwise specified herein. The slide fasteners shall be brass with a tab pull. The tape and the bead of all size fasteners (including the body opening) shall be a high temperature resistant polyamide material that has been dyed to match the color of the basic material specified in 3.4.1.1. The tape of the slide fastener shall have colorfastness in accordance with A-A-55634, except that fair colorfastness, in lieu of good colorfastness, to light will be acceptable. The finish of the chain and the other metal components shall be black chemical finish. All automatic locking sliders shall have a pin-type locking device.

3.4.5. Spacer material.

The spacer material for the bladder and bladder inflation tube shall conform to type III of MIL-DTL-43204D.

TABLE I. Slide fasteners.

GARMENT SIZE	CONTROL DRAWING	LOCATION
SR	200518653-01	Thigh take-up
SL/ML/LL/LXL	200518653-03	Thigh take-up
MR/LR	200518653-05	Thigh take-up
ALL	200518653-07	Pocket
ALL	200518653-11	Body
SR	200518653-13	Left leg
SR	200518653-15	Right leg
SL	200518653-17	Left leg
SL	200518653-21	Right leg
MR	200518653-23	Left leg

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TABLE I. Slide fasteners. – Continued

GARMENT SIZE	CONTROL DRAWING	LOCATION
MR	200518653-25	Right leg
ML	200518653-27	Left leg
ML	200518653-31	Right leg
LR	200518653-33	Left leg
LR	200518653-35	Right leg
LL	200518653-37	Left leg
LL	200518653-41	Right leg
LXL	200518653-43	Left leg
LXL	200518653-45	Right leg

3.4.6. Stiffeners.

The stiffeners for the slide fasteners, front and back body, shall be made of nylon cloth conforming to PIA-C-3953 class 2.

3.4.7. Aromatic polyamide tape.

The webbing for the body and leg socket snap tap assemblies shall be 1.50 (±0.031) inch wide conforming to PIA-T-38283 Type IV and shall approximately match the color of the base material (see 3.4.1.1). Piece-dyeing or package-dyeing is acceptable to approximately meet shade requirements for Type IV. The webbing for the leg socket snap tab assemblies may be cut on straight lines instead of curved, as called out in the patterns, to ease in manufacturing (see the example below in FIGURE 1B). NOTE: These sketches in FIGURE 1B are examples. They are not patterns and are not to scale. Do not use the sketches for manufacturing the anti-g garments.



FIGURE 1B. Example of referenced pattern cut.

3.4.8. Nylon webbing, knife pocket.

The webbing for the knife pocket reinforcement shall be untreated, conforming to Class 1, Type XII of PIA-W-4088, and approximately match the shade of the basic material in 3.4.1.1, or it shall be natural.

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3.4.9. Wrapping tape for anti-g garment hose assembly.
The tape for wrapping the anti-g garment hose assembly shall be black, 1-inch wide and conform to A-A-55809.

3.4.10. Lacer loops.
The tape for the lacer loops shall conform to MIL-W-81116, except that the sleeve may have 16 carriers, in lieu of 20 carriers; the length shall be 39.5 inches, in lieu of 45 inches for each yard of webbing; and the color shall approximately match the color of the basic material (see 3.4.1.1). Piece-dyeing or package-dyeing is acceptable to approximately meet shade requirements.

3.4.11. Thread.
Unless otherwise specified herein, the thread for all sewing operations shall be soft or bonded, 800 or 600 Denier, and 80 or 60 Tex, conforming to A-A-50195 and shall approximately match the color of the basic material as specified in 3.4.1.1. Back stitching shall be performed to prevent thread slippage.

NOTE: This thread may be used to sew the nylon mesh cloth (see 3.4.21) in lieu of the thread conforming to Type I, Class A, Tex 68-70 of A-A-59826, which is specified on AF Drawing 74204.

3.4.12. Snap fasteners.
Except for the knife pocket, the snap fasteners shall conform to NASM27980, Style 2 (socket MS27980-6B, button MS27980-1B, stud MS27980-7B, and eyelet MS27980-8B). The snap fastener for the knife pocket shall conform to NASM27983, Style 4, Construction A (button MS27983-1, socket MS27983-2, stud MS27983-3, and eyelet MS27983-4). Snaps shall be securely attached to the materials. The roller barrels in each socket and stud shall be rolled over completely and shall have a symmetrical, donut-shaped appearance with no cracks. The snap fasteners shall not rotate, or twist once installed.

3.4.13. Grommet.
The grommet shall be size 00 conforming to NASM20230-B20 and shall be oxidized a black color.

3.4.14. Fastener tape.
The hook and pile fastener tapes shall conform to A-A-55126 Type I, Class 2 or A-A-55126 Type II, Class 1 and be FR treated to be able to meet the requirements of 14 CFR Chapter 1, Section 25.853 and Appendix F, part I (a) (1) (i) 60 second Vertical Burn Test. The color shall approximately match the color of the basic material specified in 3.4.1.1. The fastener tape shall be 1 inch wide. Piece-dyeing or package-dyeing is acceptable to approximately meet shade requirements.

3.4.15. Adhesive.
The adhesive for cementing the spacer material to the coated bladder material shall be a one-part polyurethane cement (see 6.8).

3.4.16. Connector.
The connector shall conform to MIL-C-83390. This connector, deemed Inactive for New Design, is a critical component and required for use in the anti-g garment which is still being used

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throughout the DoD, including the reserves and national guard, and in support of Foreign Military Sales (FMSs) agreements.

3.4.17. Rubber sleeve.
The rubber sleeve for the connector shall be tubular and shall conform to MIL-PRF-6855/4, Class 2, Type B, Grade 60 or Grade 40. The rubber sleeve shall be 3 (±0.063) inches long, shall have an outer diameter of 0.75 (±0.063) inch, and shall have a wall thickness of 0.125 (±0.031) inch. No silicone or any other lubricant shall be applied to either the rubber sleeve or the connector prior to assembling the rubber sleeve to the connector.

3.4.18. Clamps
The clamps for securing the connector to the inflation tube and external inflation tube shall conform to Oetiker Part No. 16700031, Ref No. 25.6–706R, size 22.4 to 25.6 mm (see FIGURE 2).

3.4.19. Leather reinforcement.
The leather for the knife and leg pocket reinforcement and the reinforcement patch over the inflation sleeve opening shall be black, conforming to Type III of MIL-L-43283, except that the thickness (ounces) shall be 2 to 2.50 ounces. Leather conforming to MIL-DTL-32092, Type I, Class 1, weight 2 to 2.50 ounces is an acceptable substitute.

3.4.20. Springs for tube spacer.
The material for the springs of the tube spacer shall be a round, 0.016-inch diameter, composition 304, wire with a minimum tensile strength of 306,000 psi and a maximum of 338,000 psi per ASTM A313/A313M.

3.4.21. Cloth for the tube spacer.
The cloth for the tube spacer shall be polypropylene netting/mesh cloth with 14 x 12 strands per inch (± 1); hole size 0.05 x 0.08 (± 0.01) inches, 0.020 (± 0.010) inches thick, and 19 (± 0.5) pounds per 1000 feet squared or may be a material approved by the procuring activity before manufacture of the bladder.

3.4.22. Connector rubber band.
The connector rubber band (see FIGURE 2) shall be 0.75 (± 0.031) inch inside diameter, 1.25 (±0.063) inch long, and 0.035 inch (minimum) thickness. The rubber shall comply with MIL-PRF-6855/4 Class 2, Type A, and the color of the rubber band shall be black.

3.4.23. Webbing textile, elastic.
The elastic for the kneeboard subassembly shall conform to MIL-W-5664, Type I, Class 1 and shall be one inch wide and cut in lengths specified in TABLE VII. The color of the elastic shall approximately match the shade of the basic material in 3.4.1.1 or be black.

3.4.24. Webbing textile, nylon, kneeboard.
The webbing for the elastic kneeboard subassembly shall be 1 inch wide, cut in lengths as specified in TABLE VII, and shall conform to MIL-DTL-5038, Type II, Class 1. It shall match the shade of the basic material in 3.4.1.1.

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3.4.25. Lacing rubber band.

This rubber band is for optional use in place of the tape to secure the hank in the excess lacing cord. Any rubber band no larger than 0.25 inch wide may be used provided it does not negatively impact the form, fit, function, or appearance of the suit.

3.5. Design.

The anti-g garment shall be a wrap-around garment that extends from the body to over the calves of the wearer. It shall have an inflatable bladder that will cover the body and the leg areas of the wearer and a connector that will provide normal and emergency connection and disconnection of the bladder of the anti-g garment to and from the air pressure source. Both sides of the anti-g garment shall have external adjustment laces in the body, thigh, and calf areas; however, the adjustment laces shall have covers that can be unfastened for easy accessibility. The anti-g garment shall have a pocket on the calf of each leg, an elastic strap across the front face of each thigh, and a knife pocket on the front of the left thigh.

3.5.1. Tolerances.

Unless otherwise specified in this document, the tolerances for all measurements and stitching shall be as specified in TABLE II below. These tolerances do not apply to patterns or drawings referenced in this document.

TABLE II. Tolerance for measurements and stitching.

Specified Margin (Inches)	Positive Tolerance (Inches)	Negative Tolerance (Inches)	General Defect Classifications (Based on Deviation from Specified Margin in Inches)																
			-1	-3/4	-1/2	-3/8	-1/4	-1/8	-1/16	-1/32	0	+1/32	+1/16	+1/8	+1/4	+3/8	+1/2	+3/4	+1
≤1/16	+1/32	-1/32	CRIT	CRIT	CRIT	CRIT	CRIT	MAJ	MIN	OK	OK	OK	MIN	MAJ	CRIT	CRIT	CRIT	CRIT	CRIT
>1/16 to <2	+1/16	-1/16	CRIT	CRIT	CRIT	CRIT	MAJ	MIN	OK	OK	OK	OK	OK	MIN	MAJ	CRIT	CRIT	CRIT	CRIT
≥2 to <10	+1/8	-1/8	CRIT	CRIT	MAJ	MAJ	MIN	OK	OK	OK	OK	OK	OK	OK	MIN	MAJ	MAJ	CRIT	CRIT
≥10 to <30	+1/4	-1/4	MAJ	MAJ	MIN	MIN	OK	OK	OK	OK	OK	OK	OK	OK	OK	MIN	MIN	MAJ	MAJ
≥30 to <60	+3/8	-3/8	MAJ	MIN	MIN	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	MIN	MIN	MAJ
≥60	+1	-1	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK

At no time should any stitching or seam be less than 0.031 inches from the edge. See Table XIV for tolerance defect classifications.

3.6. Construction.

The anti-g garment needs to be constructed as specified in TABLE III; however, the manufacturer will not be required to follow the exact sequence of operations as listed therein.

3.6.1. Stitches and seams.

Stitches and seams specified in TABLE III needs to conform to ASTM D 6193. Thread breaks and ends of all stitches and seams, if not caught in other stitches or seams, needs to be securely backstitched reversing the stitching direction and sewing a continuous row of stitching superimposed on the original row of stitching for a distance of 0.50 inch. Stitching needs to be considered caught in another stitch or seam if the two stitches are in opposite directions and superimposed directly on each other for at least one inch. Thread ends of all stitches inside and outside of the anti-g garment needs to be trimmed flush with the fabric not to exceed 0.25 inch in length.

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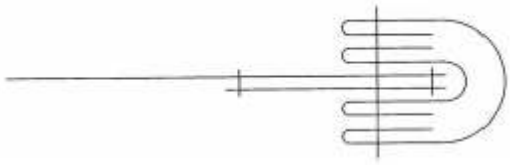
3.6.2. Bartacks.
All bartacks needs to be stitched at 40-42 stitches per inch (SPI) unless otherwise specified. The bartacks joining the spacer material and attaching the patches of the bladder (see FIGURE 3) needs to be 0.75 inch long and centered. All other bartacks needs to be 0.50 inch long unless otherwise specified.

TABLE III. Manufacturing operations.

No.	Description of Operation	Stitch Type	Seam & Stitching Type	Stitches per Inch
1.	**Cut parts (see 3.11.1).** Cut the parts (including the bladders) of the anti-g garment in accordance with the patterns which show shape, size, directional lines, and marking for the proper assembly of the parts. NOTE: Trimming of the body lacer covers, the thigh lacer covers, and the calf lacer covers will be permitted as specified in Operations 5.b, 6.b, and 6.e.			
2.	**Shade mark.** Mark all parts to ensure a uniform shade throughout the anti-g garment.			
3.	**Sear raw edges.** Sear the raw edges of all parts made of nylon material.			
4.	**Sew loops of lacer loop tape.** Sew one row of stitching along each length of the lacer loop tape, through the base of each loop to prevent the loops from slipping.	301	SSa-1	10-12
5.	**Make body lace covers.** a. Select the length of the fastener tapes for the body lacer covers in accordance with TABLE IV.			

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TABLE III. Manufacturing operations. – Continued

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
5.	**Make body lace covers – contd.**			
	b. Trim the fastener tape, if necessary, to fit the body lacer covers at the top and the bottom. Turn each body lacer cover under 0.375 inch. Sew the hook fastener tape to each body lacer cover, at the pattern location, with a single row of stitching 0.125 (+0.063 or -0) inch from the edge on all four sides. NOTE: Operation 5.b shall be performed before Operation 5.c so that the fastener tape will be sewn to the body lacer covers before the body lacer covers are bound.	301	LSd-1	8-10
	c. Bind each body lacer cover at each end with the finished binding 0.063 (+ 0.063 or -0.031) inch from the edge. (See instructions for preparing the binding in 3.4.2.) Before completing the stitching of the binding, at the top and the bottom of each lacer cover, fold the raw edges of each end of the binding (two ends per lacer cover) 0.50 (± 0.125) inch so that no raw edges are exposed (see turned-under binding in the sketch below). Complete the stitching of the binding to the fastener tape. 	301	BSc-1	10-12
6.	**Make leg (thigh and calf) lacer covers.**			
	a. Select the length of the fastener tapes for the thigh lacer covers in accordance with TABLE IV.			

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TABLE III. Manufacturing operations. – Continued

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
6.	**Make leg (thigh and calf) lacer covers – contd.**			
	b. Trim the fastener tape, if necessary, to fit the thigh lacer covers at the top and the bottom. Turn each thigh lacer cover under 0.375 inch. Sew the hook fastener tape to each thigh lacer cover, at the pattern location, with a single row of stitching 0.125 (+ 0.063 or - 0) inch from the edge on all four sides. NOTE: Operation 6.b shall be performed before Operation 6.c so that the fastener tape will be sewn to the thigh lacer covers before the thigh lacer covers are bound.	301	LSd-1	8-10
	c. Bind each thigh lacer cover at each end with the finished binding 0.063" (+ 0.063 or - 0.031) inch from the edge. (See instructions for preparing the binding in 3.4.2.) Before completing the stitching of the binding, at the top and the bottom of each lacer cover, fold the raw edges of each end of the binding (two ends per lacer cover) 0.50 (± 0.125) inch so that no raw edges are exposed (see turned-under binding sketch in Operation 5.c above). Complete the stitching of the binding to the fastener tape.	301	BSc-1	10-12
	d. Select the length of the fastener tapes for the calf lacer covers in accordance with TABLE IV.			
	e. Trim the fastener tape, if necessary, to fit the calf lacer covers at the top and the bottom. Turn each calf lacer cover under 0.375 inch. Sew the hook fastener tape to each calf lacer cover, at the pattern location, with a single row of stitching 0.125 (+ 0.063 or - 0) inch from the edge on all four sides. NOTE: Operation 6.e shall be performed before Operation 6.f so that the fastener tape will be sewn to the calf lacer covers before the calf lacer covers are bound.	301	LSd-1	8-10

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TABLE III. Manufacturing operations. – Continued

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
6.	**Make leg (thigh and calf) lacer covers – contd.**			
	f. Bind each calf lacer cover at each end with the finished binding 0.063 (+ 0.063 or - 0.031) from the edge (See instructions for preparing the binding in 3.4.2). Before completing the stitching of the binding, at the top and the bottom of each lacer cover, fold the raw edges of each end of the binding (two ends per lacer cover) 0.50 (± 0.125) inch so that no raw edges are exposed (see turned-under binding sketch in Operation 5.c above). Complete the stitching of the binding to the fastener tape.	301	BSc-1	10-12
7.	**Make thigh take-ups.**			
	a. Select the slide fasteners for the thigh take-ups in accordance with TABLE I. Zigzag stitch the tapes of the slide fasteners together from the bottom stop to the ends of the tapes.	304	SSa-1	20-26
	b. Sew the 0.75 inch wide stiffener to the inside of the thigh take-up fly, right and left, on the long center line from end to end (one full stitch over each end of the stiffener) with a single row of stitching.	301	SSa-1	10-12
	c. With the open end of the slide fastener at the top, center the top and bottom stops between the top and bottom of the thigh take-ups. Match the edges of the slide fastener tape with the edges of the thigh take-ups fly. Cut the ends of the slide fastener tapes off, even with the cover, and sew all around with a single row of stitching 0.125 (± 0.063) inch from the edge.	301	SSa-1	4-6
	d. Bind all around the leg take-ups, catching the ends of the slide fastener tapes in the binding, with the finished binding 0.063 (+ 0.063 or - 0.031) inch from the edge (see instructions for preparing the binding in 3.4.2).	301	BSc-1	10-12

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TABLE III. Manufacturing operations. - Continued

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
7.	**Make thigh take-ups. – contd.**			
	e. Cut two 1.25 inch by 6 inch pieces from the basic fabric cut on a 45 degree bias as specified in paragraph 3.4.1.1. Fold the edges to center and then in half to form a 0.313 inch wide zipper thong. Sew around the perimeter 0.063 inch from the edge.	301	SSc-1	10-12
7.	**Make thigh take-ups – contd.**			
	f. Thread the 6 inch piece through the pull of each slide fastener. Fold the webbing in the center. Fold the raw ends under 0.50 inch and bartack. Sew another bartack close to the pull.	bartack		40-42
8.	**Prepare slide fasteners for body and leg openings.**			
	a. Select the slide fasteners for the body and the leg openings in accordance with TABLE I.			
	b. Prefold the slide fastener tapes for the leg openings at the open ends with a triangular fold so that the beaded edge of the tape is across the end and the fold is to the inside of the case. Sew with a single row of stitching 0.094 (+0.031 or - 0.063) inch from the edge. Prefold and sew the slide fastener tape for the body opening in the same manner as for the leg openings, making certain that the fold is placed so that the length of the slide fastener tape will be 8.625 (± 0.125) inches.	301	EFa-1	10-12
	c. Cut two 1.25 inch by 6 inch pieces from the basic fabric cut on a 45 degree bias as specified in paragraph 3.4.1.1. Fold the edges to center and then in half to form a 0.313 inch wide zipper thong. Sew around the perimeter 0.063 inch from the edge.	301	SSc-1	10-12
	d. Thread the 6 inch piece through the pull of each slide fastener. Fold the webbing in the center. Fold the raw ends under 0.50 inch and bartack. Sew another bartack close to the pull.	bartack		40-42

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TABLE III. Manufacturing operations. – Continued

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
8.	**Prepare slide fasteners for body and leg openings – contd.**			
	e. Cut three 1.25 inch by 7 inch pieces from the basic fabric cut on a 45 degree bias as specified in paragraph 3.4.1.1. Fold the edges to center and then in half to form a 0.313 inch wide strip. Sew around the perimeter 0.063 inch from the edge. Fold the tape in the center. Fold the raw ends under 0.50 inch. Join, forming a box and "X" with a single row of stitching 0.063 inch from the edge to the separating end of the fastener slide opposite the slider, with the tabs down 0.625 inch from the end of the slide fastener tape. Sew the tape to the inside of the anti-g garment.	301	SSa-1	10-12
	f. Sew the loops of the two tapes on the leg opening slide fasteners closed by sewing both sides of the loops with a single row of stitching 0.063 inch from the edges.	301		10-12
9.	**Make body protective cover.**			
	a. Position a 1-inch square reinforcement of the basic fabric as marked on the pattern so that the center of the reinforcement will be 0.75 inch down from the top of the finished fly on the inside and sew with a single row of stitching 0.125 inch from the edge. Position another 1-inch square reinforcement of the basic fabric so that the center of the reinforcement will be 5.50 inches below the center of the upper reinforcement and equidistant from the edge of the protective fly cover and sew with a single row of stitching 0.125 inch from the edge.	301	SSa-1	10-12
	b. Fold the body protective fly cover lengthwise on the fold line, the material face to face; sew across the bottom and the top with a single row of stitching 0.375 inch from the edge.	301	SSe-2(a)	10-12

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TABLE III. Manufacturing operations. – Continued

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
9.	**Make body protective cover. – contd.**			
	c. Turn and insert the stiffener, folding the raw edge to the inside enclosing the stiffener. Sew all around the top, bottom, and two sides with a single row of stitching, 0.125 inch from the edge. Sew four additional rows of stitching, evenly spaced end to end.	301	SSe-2(b)	10-12
	d. Sew two boxes, "X" 0.75 by 0.75 inch, 0.125 inch from the edge of two reinforcements described in Operation 9.a with a single row of stitching.	301	SSa-1	10-12
	e. Punch a hole, diameter 0.125 inch, in the center of each box "X" described in Operation 9.d. Install stud NASM 27980-7B and eyelet NASM 27980-8B through the hole at the center of box "X" stitch, ensuring the stud is on the side that matches with socket of tab assembly.			
	f. Attach the slide fastener tape (with the slider attached) to the back edge of the body protective fly, the open end at the top; sew with a single row of stitching 0.125 inch from the edge. The slide fastener will extend below the bottom edge of the body protective fly by 0.625 (± 0.125) inch.			

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
10.	**Make leg protective cover.**			
	a. Position a 1-inch square box reinforcement of the basic fabric, as marked on the patterns, so that the center of the reinforcement will be 2.625 inches down from the top of the leg protective fly finished edge and on the inside. Sew with a single row of stitching 0.125 inch from the edge. Position another 1-inch square box reinforcement of the basic fabric so that the center of the reinforcement will be 1 (± 0.125) inch up from the bottom edge of the legs and 0.625 (± 0.125) inch from the outside edge of the protective fly. Sew with a single row of stitching 0.125 inch from the edge.	301	SSa-1	10-12
	b. Fold the leg protective fly cover lengthwise on the fold line, the face sides of the material together; sew across the bottom and the top with a single row of stitching 0.375 inch from the edge.	301	SSe-2(a)	10-12
	c. Turn and insert the stiffener, folding the raw edges to the inside enclosing the stiffener; sew all around the top, the bottom, and the two sides with a single row of stitching 0.125 inch from the edge. Sew four additional single rows of stitching evenly spaced from end to end.	301	SSe-2(b)	10-12
	d. Sew a box “X” 0.75 by 0.75 inch, 0.125 inch from the edge of reinforcement described in Operation 10.a with a single row of stitching.	301	SSa-1	10-12
	e. Punch 0.125 inch diameter hole in the center of each box “X” described in Operation 9.d. Install stud NASM 27980-7B and eyelet NASM 27980-8B through the hole at center of box “X” stitch, ensuring the stud is on the side that matches with socket of tab assembly.			

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
10.	**Make leg protective cover. – contd.**			
	f. Attach the slide fastener tape (with the slider attached) to the back edge of the leg protective fly, the open end at the bottom; sew with a single row of stitching 0.125 inch from the edge.	301	SSa-1	10-12
11.	**Assemble pockets.**			
	a. Bind the side edges of the pockets with the finished binding 0.063 (+ 0.063 or - 0.031) inch from the edge (see instructions for preparing the binding in 3.4.2).	301	BSc-1	10-12
	b. Position leather patch on pocket face (front and bottom of pocket) aligning cut edge of leather with edge of pocket pattern, 0.50 inch below bottom edge of zipper opening (see FIGURE 14). Stitch leather to pocket with a single row of stitching 0.125 (±0.031) inch from cut edge of leather.	301	SSa-2	9-11
	c. Place the slide fastener at the marked location on the pocket to the inside; join with a single row of stitching 0.063 inch from the edge all around. (The bridge end shall be at the lowest point on the pocket opening.)	301	SSa-1	10-12
	d. Slit through the center and diagonally at the ends. Turn the fabric under 0.25 inch; sew with a single row of stitching 0.063 inch from the folded edge.	301	LSd-1	10-12
	e. Sew the bottom corners, the face sides of the material together, with a single row of stitching 0.125 inch from the edge.	301	SSa-1	10-12
	f. Raise the sides and the bottom edge of the pockets, at the pattern marks, with a single row of stitching 0.125 inch from the folded edge.	301	OSf-1	10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
11.	**Assemble pockets. – contd.**			
	g. Cut two 1.25 inch by 6 inch pieces from the basic fabric cut on a 45 degree bias as specified in paragraph 3.4.1.1. Fold the edges to center and then in half to form a 0.313 inch wide zipper thong. Sew around the perimeter 0.063 inch from the edge.	301	SSc-1	10-12
	h. Thread the 6 inch piece through the pull of each slide fastener. Fold the strip in the center. Fold the raw ends under 0.50 inch and bartack. Sew another bartack close to the pull.	bartack		40-42
	i. Cut four, 2.75-inch squares of the basic fabric to reinforce the slide fasteners. Turn the outside edges of each reinforcement under 0.25 inch on all four sides, and stitch two reinforcements 0.063 inch from the edge to each pocket with a box “X” stitch inside a box through all thicknesses of fabric and zipper tape. This box “X” may be sewn on top of the top edge of the leather. The reinforcements shall be positioned so that a reinforcement will be adjacent to each end of the metal portion of the slide fastener (centered from the top and the bottom on the inside of the pocket).	301	LSa-1	10-12
12.	**Make leg loop assembly.**			
	a. Cut two pieces of nylon tape in accordance with the patterns, fold at the center, sew a box “X” 0.75 by 1.25 inches, 0.125 inch from the folded edge. Punch a 0.125 inch diameter hole in the center of box “X” stitch, install button NASM 27980-1B and socket NASM 27980-6B through the hole at center of box “X” stitch.	301	SSa-1	10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
13.	**Make body loop tab assemblies.**			
	a. Cut two pieces of nylon tape in accordance with the patterns, fold on the line, sew a box "X" 0.75 by 1.25 inch, 0.125 inch from the folded edge. Punch a hole, diameter 0.125 inch, in the center of box "X" stitch, install button NASM27980-1B and socket NASM27980-6B through the hole at the center of box "X" stitch.	301	SSa-1	10-12
	b. Fold the 0.375 inch extended end over the short end. Sew all around the tab with a single row of stitching 0.063 inch from the edge.	301	Similar to SS1-1	10-12
14.	**Construct bladder.**			
	a. The polyurethane coated cloth (see 3.4.1.2) specified for the bladder shall be cut in lengths required by the applicable patterns. The bottom half of the bladder, with the coated surface face up, shall be laid on a clear flat surface.			
	b. Four strips of the spacer material (see 3.4.5) shall be cut in lengths required by the applicable pattern, except that the horizontal spacer extending across the inside of the abdominal bladder shall have 2 to 3 inches of slack (the length of spacer between the two patches in the abdominal bladder shall be 2 to 3 inches longer than the distance, in the flat, between the two patches).The width of the spacer in the bladder extension and the abdominal bladder shall be 0.75 inch. The width of the leg spacers shall be 1 inch.			

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
14.	**Construct bladder. – contd.**			
	c. The ends of the legs and abdominal spacer material shall be overlapped and joined to patches that are made of polyurethane coated nylon cloth (see 3.4.1.2) and that are 1 inch by 1.75 inches by bartacking as shown on FIGURE 3. Two additional pieces of spacer material, 1 inch wide by 10 inches long, shall be sewn to the upper portion of the leg spacer material (one length per leg).	bartack		40-42
	d. The upper end of the piece of spacer material that is 1 inch by 10 inches shall be placed 1 inch below the patch joining the leg spacer to the abdominal spacer and shall be sewn to the leg spacer with two rows of stitching 0.25 inch from each edge of the leg spacer with the thread specified (see 3.4.11) and all ends backstitched a minimum of 1 inch. The bladder extension spacer material shall be overlapped on the abdominal spacer material 2.50 inches from the joint between the leg and abdominal spacer material and shall be similarly bartacked (no patch).	301	LSa-2	6-8
	e. The spacer material (anti-block system) shall be positioned on the coated side of the bottom half of the bladder as indicated on the applicable pattern and then secured by Radio Frequency (RF) welding or cementing patches to the bottom of the bladder as shown on FIGURE 4.			
	f. The bladder halves shall be positioned, and RF welded together so that the bonded area will be 0.125 (±0.031) inch wide. The distance from the outside edge of the weld to the edge of the bladder material shall not be greater than 0.50 inch nor less than 0.063 inch. The bonded seam shall be straight, continuous, and parallel to the edge of the bladder.			

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
14.	**Construct bladder. – contd.** g. No part of the spacer material or the attachment patches shall be caught in any portion of the welded seam. The completed bladder shall be inflated to 2 pounds per square inch (psi) air pressure. The seams of the inflated bladder shall be inspected for proper joining and adhesion. The distance from the outside edge of the weld to the edge of the bladder material shall not be greater than 0.50 inch nor less than 0.063 inch, not including the bladder tabs and points on the pattern. The upper tab of the finished bladder shall be stamped with the abbreviated anti-g garment size.			
15.	**Hem body bladder cover.**			
	a. Hem the upper edge of the separate leg extension to the inside, 0.25 inch, with a single row of stitching 0.063 inch from the edge.	301	EFa-1	10-12
	b. Fold the end of the outside body bladder cover leg, to which the separate leg extension is to be attached, to the inside 0.25 inch and hem with a single row of stitching 0.063 inch from the edge. NOTE: The terms “back body bladder cover,” “front body bladder case,” “outside cover,” and “forward cover” as indicated in this document and on the pattern drawings of the bladder cover shall be interpreted as that portion of the bladder cover on the outside (away from the wearer’s body) and has a truncated section cut off. The term “back body bladder cover” or “front body bladder case” shall be interpreted as the cover positioned next to the wearer’s body and is full size as indicated on the pattern drawings.	301	EFa-1	10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
15.	**Hem body bladder cover. – contd.**			
	c. Lap the hemmed end of the separate leg extension up under the body bladder cover leg end until the lower edge is even with the opposite leg end and sew at the sides with a single row of stitches 0.125 inch from the edge.	301	SSa-1	10-12
16.	**Join bladder case.**			
	a. Join the front and the back body bladder case, material face to face, with a single row of stitching 0.25 inch from the edge, leaving the legs, the bladder inflation sleeve, and one inch at tab locations (top, bottom, and right side) open.	301	SSe-2(a)	10-12
	b. Turn and topstitch with a single row of stitching 0.063 inch from the edge.	301	SSe2(b)	10-12
17.	**Hem leg bladder covers.**			
	a. Hem the leg bladder covers to the inside, 0.25 inch, down the sides and around the bottom and the top, with a single row of stitching 0.063 inch from the edge.	301	EFa-1	10-12
18.	**Make body section.**			
	a. Join the left side of the body front to the left side of the body back with a double row of stitching, 0.063 inch from the edges, 0.25 inch gage.	301	(see FIGURE 5)	8-10
	b. Sew a 1.125 inch wide bias-cut piece of the basic fabric folded to 0.625 inch over the center of the body front and back body seam (inside) with a single row of stitching, 0.063 inch from the edge, each side of the binding.	301	LSd-1	10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
18.	**Make body section. – contd.**			
	c. Center the bias-cut pieces of the basic fabric on the stiffeners. Sew three rows of stitching from end to end with one row in the center and one row on the right and the left sides of the center stitching 0.125 inch.	301	SSa-1	10-12
	d. Place the stiffeners at the pattern marks on the back body and front body and stitch a single row of stitching, 0.063 inch from the edges, each side of the binding.	301	SSa-1	10-12
	e. Place the four lacer loop tapes on the body back, in accordance with the pattern marks, the ends even with the top and bottom edges of the body back, loops facing loops, and join with a double row of stitching 0.063 inch from the edge 0.188 inch gage.	301	SSa-1	10-12
	f. Bartack both ends of lacer loop across full width of the lacer loop including the loop material, 0.375 inch from the top and bottom edge.	bartack		40-42
	g. Place the pile fastener tape on the lacer loop tape on the body back, 0.125 inch from the loops, and join with a single row of stitching 0.125 (+ 0.063 or - 0) inch from the edge.	301	SSa-1	8-10
	h. Bind across the top, catching the seam tape and the lacer loop tape with the finished binding 0.063 (+ 0.063 or - 0.031 from the edge (see instructions for preparing the binding in 3.4.2).	301	BSc-1	10-12
	i. Place a blank label (5 inches long by 1.50 inches wide) vertically and centered on the inside, center back section between the lower stiffeners. Sew label to the back section with a single row stitching 0.125inch from the edge on all four sides (see FIGURES 13A and 13B).	301	SSa-1	8-10

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
18.	**Make body section. – contd.**			
	j. Place the size, care, and manufacturer labels on the inside of the center back centered between the two bindings with a single row of stitching 0.125 inch from the edge of all four sides as shown in FIGURES 13A and 13B. Place the Item Unique Identification (IUID) label horizontally centered on the inside of the center back between the body snap tabs with a single row of stitching 0.125 inch from the edge on all four sides (see FIGURES 13A and 13B).	301	SSa-1	8-10
	k. Bind across the bottom of the body front and body back with the finished binding 0.063 (+ 0.063 or - 0.031) inch from the edge (see instructions for preparing the binding in 3.4.2).	301	BSc-1	10-12
	l. Cut a 1.25 inch by 4.75 inch piece from the basic fabric cut on a 45 degree bias as specified in paragraph 3.4.1.1. Fold the edges to center and then in half to form a 0.313 inch wide hanger loop. Sew around the perimeter 0.063 inch from the edge.	301	SSc-1	10-12
	m. Fold the raw end under 0.375 inch and bartack one end 0.75 inch down from the top of the edge binding-on-binding strip over the seam joining the body front to the side back. Fold the binding to the inside; bartack the other end 1.75 inches from the top edge binding	bartack		40-42
	n. Turn the body lacer cover under 0.375 inch. Place the finished edge of the body lacer cover on the inner row of lacer loop tape, edge to edge; join with a double row of stitching 0.063 inch from the edge, 0.188 inch gage.	301	LSa-2	10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
19.	**Attach inflation tube opening reinforcement.**			
	a. Lay the 3.313 inch square reinforcement piece of the basic fabric at the marked location on the outside of the body front, evenly spaced over the hole position. Sew around the hole marking, 1.125 inch diameter, with a single row of stitching.	301	SSe-2(a)	10-12
	b. Trim the 0.25 inch inside stitching and make six small cuts around the edge of the hole but not closer than 0.063 inch from the stitching. Turn the reinforcement piece through the opening; topstitch with a single row of stitching 0.063 inch from the topstitched edge.	301	SSe-2(b)	10-12
	c. Turn the outside edge of the reinforcement under 0.50 inch all around and sew to the body front with a single row of stitching 0.063 inch from the folded edge.	301	LSd-1	10-12
	d. Lay the octagon-shaped leather reinforcement piece evenly spaced over the hole position, on the inside of the body front, and sew around the hole 0.094 inch from the edge (on the inside). Sew around the outside edge of the leather patch 0.094 inch from the edge (on the inside).	301	LSa-2	9-11
20.	**Attach body bladder cover.**			
	a. Place the assembled body bladder cover on the body front to the inside at the marked location with the bladder inflation sleeve on the left side; join with a double row of stitching 0.063 (+ 0.063 or - 0.031) inch from the edge, 0.063 inch gage. Leave open the end of the inflation tube external covering, the leg openings, and 1 inch at the tab location, top, bottom, and right side.	301	SSa-2	10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
21.	**Attach assembled body protective fly.**			
	a. Turn the edge of the cloth under 0.375 inch on the body front. Sew the slide fastener tape side with the protective fly attached; with 0.25 inch of the slide fastener tape exposed; with a double row of stitching 0.063 inch from the folded edge 0.25 inch gage. The slide fastener shall be on the inside with the opening on the top and the closed end extending below the body panel by 1 (± 0.125) inch.	301	LSd-2	10-12
	b. Sew the upper body button socket tab to the inside of the side back (front) on the right-hand side of the slide fastener edge as marked on the patterns, forming a box and "X" stitch, 0.375 inch by 1.375 inches, 0.063 inch from the edge. The upper button socket tab shall match the stud eyelet on the body protective fly. Sew the lower button, socket tab 5-0.50 inches (center line to center line) below the upper button socket tab. The lower button socket tab shall match the lower stud eyelet on the body protective fly.	301	LSd-1	10-12
	c. Turn the edge of the cloth under 0.375 inch on the body front. Sew the slide fastener tape side without the fly attached, with ¼ inch of the slide fastener tape exposed, with a double row of stitching 0.063 inch from the folded edge ¼ inch gage. The slide fastener shall be on the inside with the opening on the top and the closed end extending below the body panel by 1 ± 0.125 inch.	301	LSd-2	10-12
22.	**Make leg sections.**			
	a. Sew a 1.125 inch wide bias-cut piece of the basic fabric folded to 0.625 inch on the inside of the legs, at the positions marked at the sides of the pocket and at the leg take-ups, with a single row of stitching 0.063 inch from the edge of each side of the binding, right and left.	301	LSd-1	10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
22.	**Make leg sections. – contd.**			
	b. Turn the neck of the pockets inside 0.25 inch; sew across the knees at the marked locations with a double row of stitching 0.063 inch from the edge, 0.063 inch gage, leaving 0.50 inch at each edge open, the pocket to the outside, the slide fastener closing to the outside.	301	LSd-2	10-12
	c. Place the webbing, with the button socket assembly (NASM27980-1B, -6B) attached to the webbing, on the inside of the legs. The top edge of the webbing shall be parallel with and down from the top, 1.75 inches. Sew across the back end of the webbing, following the contour of the bladder cover locations, with a single row of stitching. The socket of the button socket tab shall face the body and shall match the stud on the leg protective fly.	301	LSd-1	10-12
23.	**Make and attach knife pocket.**			
	a. Lay the knife pocket reinforcement piece on the pocket fabric, face up, in accordance with the notches and the drill marks. Sew the knife pocket reinforcement piece to the pocket with a single row of stitching 0.094 (± 0.031) inch from the edge on all four sides	301	SSa-1	10-12
	b. Fold the pocket fabric lengthwise with the raw edges even. Bind the square end with the finished binding 0.063 (+ 0.063 or - 0.031) inch from the edge (see instructions for preparing the binding in 3.4.2). The knife pocket reinforcement shall be on the outside after folding.	301	BSc-1	10-12
	c. Bind the straight edge of the small flap reinforcement piece for the top pocket with the finished binding 0.063 (+ 0.063 or - 0.031) inch from the edge (see instructions for preparing the binding in 3.4.2).	301	BSc-1	10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
23.	**Make and attach knife pocket. – contd.**			
	d. Position the flap reinforcement piece in accordance with the notch and the drill marks at the round end of the pocket with the flap piece on the same side as the webbing piece. Sew the flap piece to the two top plies of the pocket fabric with a single row of stitching 0.125 inch from the edge.	301	SSa-1	10-12
	e. Place the leather reinforcement piece on the outside of the knife pocket, flat and smooth, with the bottom edge resting on the center notch on the knife pocket and the top edge toward the round end of the knife pocket. Stitch the leather reinforcement piece 0.125 inch from the edge on all four sides. Double stitch across the top of the leather reinforcement piece with 0.25-inch gage.	301	SSa-1	9-11
	f. Fold the pocket fabric on the center notch. The leather reinforcement piece shall be on the outside after binding. The tape shall extend 0.50 inch beyond the end on each side for turning under.	301	BSc-1	10-12
	g. Place a single row of stitching across the flap through both plies at the round end of the flap, 1 inch from the open bound end.	301	SSa-1	10-12
	h. Install a size 00 grommet on the pocket, between the square end and the round end of the flap reinforcement piece, in accordance with the drill marks.			

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operations	Stitch Type	Seam & Stitch Type	Stitches per Inch
23.	**Make and attach knife pocket – contd.**			
	i. Position the reinforcement patch on the left thigh front so that the knife pocket is just inside the bladder case stitch line at the front, bottom, and top corners and so that the side of the pocket is parallel to the leg opening slide fastener. Turn the raw edges under 0.375 inch at the bottom and the sides; join with a double row of stitching 0.063 inch from the folded edge 0.25-inch gage. Trim the top even with the top of the thigh.	301	LSd-2	10-12
	j. Position the knife pocket on the reinforcement patch, with the round end up and the opening on the inside, in accordance with the drill marks. Start at the bound opening and sew down the side, across the bottom, and up to the bound opening, turning the bias end of the tape under on each side and backstitching 0.50 inch on each side, with a double row of stitching 0.063 inch from the edge, 0.125 inch gage. The distance between the inside longitudinal stitch lines shall be 2.25 inches.	301	LSd-2	10-12
	k. Sew across the bound square end at the top of the pocket opening, securely tacking the top of the pocket to the patch and the anti-g garment with a single row of stitching.	301	SSa-1	10-12
	l. Stamp one snap fastener socket and button to the center of the flap so that the dot on the snap fastener will be at the top of the flap in accordance with the drill marks at the rounded end.			

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
23.	**Make and attach knife pocket – contd.** m. Stamp one snap fastener stud and eyelet, reinforced with bladder material on the inside, in the anti-g garment to correspond to the snap fastener socket. If the stud and eyelet location is on top of the binding around the thigh opening, relocate position just below the binding. Snaps shall be securely attached to the materials. Ensure that roller barrels in each socket and stud shall be rolled over completely and shall have a symmetrical donut shaped appearance with no cracks. n. Thread a 67 (± 3) inch long lanyard through the grommet. Tie the lanyard with a bowline knot. Sear both ends of the lanyard. Fold the lanyard in layers that are 4 inches long. Position the folded lanyard in the pocket, parallel with the pocket, with all of the folds within the pocket. Close the snap fastener.			
24.	**Attach leg bladder covers.** a. Place the leg bladder covers on the legs at the marked locations on the patterns. Join down each side, leaving the top and 1 inch at the marked position for the tabs open, with a double row of stitching 0.063 (+ 0.063 or - 0.031) inch from the edge, 0.063-inch gage. Fold back 0.25 inch at the neck of the pocket left loose so that the pocket is not stitched through in this operation.	301	LSd-2	10-12
25.	**Attach pockets.** a. Sew the pockets to the lower legs on the center of the basic fabric bias binding, down each side, with a single row of stitching 0.063 inch from the edge of the pocket binding. b. Join the pockets at the bottom and the upper edges at the knee opening to the legs with a single row of stitching 0.125 inch from the edge.	301 301	LSd-1 SSa-1	10-12 10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
26.	**Bind knee openings and bottom of legs.**			
	a. Bind the knee openings (including the pockets) and the bottom of the legs (including the pockets and the lower edge of the leather patches) with the finished binding 0.063 (+ 0.063 or - 0.031) inch from the edge (see instructions for preparing the binding in 3.4.2).	301	BSc-1	10-12
27.	**Join lacer loop tapes to the thighs and calves.**			
	a. Place the eight lacer loop tapes the length of each thigh and each calf, in accordance with the marks on the patterns, with the ends even with the top and the bottom edge, with loops facing loops. Join the lacer loop tapes to each thigh and each calf with a double row of stitching 0.063 inch from the edge 0.188 inch gage.	301	SSa-2	9-11
	b. Bartack both ends of each lacer loop, across the full width of the lacer loop including the loop material, 0.375 inch from the top and bottom edge.	bartack		40-42
28.	**Join pile fastener tape to thighs and calves.**			
	a. Place the pile fastener tape on the lacer loop tape on each thigh and each calf, 0.125 inch from the loops, and join with a single row of stitching 0.125 (+ 0.063 or - 0) inch from the edge.	301	SSa-1	8-10
29.	**Attach assembled leg (thigh and calf) lacer covers.**			
	a. Turn the raw edge of each assembled leg (thigh and calf) lacer cover under 0.375 inch; place the finished edge of each leg lacer cover on the inner row of the lacer loop tape, edge to edge. Join with a double row of stitching 0.063 inch from the edge, 0.188-inch gage.	301	LSa-2	10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
30.	**Make elastic kneeboard subassembly.** NOTE: The respective webbing ends of the elastic kneeboard assembly shall be trimmed, if necessary, so that the elastic kneeboard subassembly lays flat across the thigh cover. a. Sear-cut webbing (MIL-PRF-5038 Type II, Class 1, 1 inch wide) 2.75 inches long. Cut 1 inch wide elastic ((MIL-W-5664, Type I, Class 1) in accordance with TABLE VII.			
	b. Sew elastic to the webbing in accordance with FIGURE 6 and FIGURE 7. Repeat for remaining end.	301	SSa-1	8-10
	c. Position one end of the kneeboard elastic subassembly (face up) 5 inches from top of and perpendicular to each respective thigh take-up assembly in accordance with FIGURE 6 and attach to the anti-g garment in accordance with FIGURE 8.	301	SSa-1	8-10
	d. Position the remaining free end of the kneeboard elastic subassembly (face-up) to the opposite side of the leg section square with the thigh-take up attachment point in accordance with FIGURE 6 and attach to the anti-g garment in accordance with FIGURE 9.			

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
31.	**Join leg take-ups.** a. Position one end of the kneeboard elastic subassembly (face up) on top of or under the thigh take-up assembly in accordance with pattern markings. Join the assembled right and left take-ups to the outside at the thigh, in accordance with the pattern marks, down the sides and across the top with a double row of stitching 0.063 inch from the edge, 0.188-inch gage. The slide fasteners shall close at the top. NOTE: Use proper thread tension and fabric tension to minimize waviness (chain flatness of slide fastener).	301	LSa-2	10-12
32.	**Sew around reinforcement.** a. Sew around the perimeter on the edges of the reinforcement inside of the leg take-ups with a single row of stitching, 0.125 (± 0.063) inch from the top and side edges, 0.25 (± 0.063) inch from the bottom edge (end closest to the slider when the zipper is in the full open position).	301	SSa-1	10-12
33.	**Attach assembled leg protective flies.** Position the remaining end of the kneeboard elastic subassembly (face-up) on the leg section in accordance with pattern markings. Turn the edge of the cloth and elastic subassembly under 0.375 inch. Place the edge of the cloth down the legs on the slide fastener tapes so that the separating ends of the slide fastener will be at the top, the protective fly will open to the back, the corners at the knee holes will be spaced 0.75 (± 0.125) inch apart on the slide fastener tapes, and 0.25 inch of the slide fastener tape will be exposed. Join with a double row of stitching 0.063 inch from the edge, 0.25-inch gage.	301	LSd-2	10-12

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
34.	**Install ankle snap fasteners.** a. Sew a box "X" 0.75 by 0.064 inch on the lower edge of the legs to the side opposite the protective fly, 0.875 inch from the bottom edge and 0.625 inch from the edge of the slide fastener teeth, aligned with the ankle snap stud installed in Operation 10. b. Punch 0.125 inch diameter hole in the center of the box "X" stitch described in Operation 34a. c. Install button NASM27980-1B and socket NASM27980-6B through the hole at the center of the box "X" stitch with the button on the face side of the cloth. Snaps shall be securely attached to the materials. NOTE: The slide fastener must function properly as designed, and the snap cannot come unsnapped or cause bulging at any time when the slide fastener is closed.			
35.	**Join upper ends of legs to body front.** a. Join the upper ends of the legs to the body front, with the face sides of the material together, in accordance with the marks on the patterns, with a single row of stitching 0.375 inch from the edge.	301	LSq-2(a)	10-12
35.	**Join upper ends of legs to body front. – contd.** b. Turn and topstitch with a single row of stitching 0.063 inch from the folded edge. c. Sew an additional row of stitching 0.313 inch from the folded edge.	301 301	LSq-2(b) SSa-1	10-12 10-12
36.	**Insert adjustment laces in take-ups.** a. Select the adjustment laces in accordance with TABLE V.			

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
36.	**Insert adjustment laces in take-ups. – contd.** b. Insert the adjustment laces into the lacer loops for the body take-ups as follows: Start lacing from the bottom, similar to lacing shoes. Route the adjustment laces through each lacer loop on each side (loose ends exiting from the top lacer loops). Leave the lacing in the full, open position. Fold the excess lacing into a hank that is 3 inches long. Wrap the hank with one to two turns of 2-inch wide tape that conforms to SAE AMS-T-22085. NOTE: For all lacing, each available loop on each side shall be threaded. If the number of loops is uneven, only one or two topmost loop(s) per assembly may be left open. The open loop(s) must be on the same side of the lacing. An available loop is any loop that the lacing needle can fit through. All lacing hanks may be secured with a rubber band (3.4.25) in place of the 2-inch wide tape. c. Insert the adjustment laces into the lacer loops for the thigh take-ups in the same manner as specified for the body take-ups in Operation 36.b, except that the lacing shall start from the top lacer loops and exit from the bottom lacer loops. Leave the lacing in the full, open position. Fold the excess lacing into a hank that is 3 inches long. Wrap the hank with one to two turns of 2 inch wide tape that conforms to SAE AMS-T-22085. d. Insert the adjustment laces into the lacer loops for the calf take-ups in the same manner as specified for the body take-ups in Operation 36.b. Leave the lacing in the full, open position. Fold the excess lacing into a hank that is 3 inches long. Wrap the hank with one to two turns of 2-inch wide tape that conforms to SAE AMS-T-22085.			

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
37.	**Final assembly.** Install the bladder through the openings at the top of the legs left open for this purpose and into the bladder covers. Insert the bladder tabs through the 1 (+ 0.125 or - 0) inch wide openings; sew the tabs with a double row of stitching 0.063 inch from the edge, 0.063 inch gauge.	301	SSa-2	10-12
38.	**Attach connector assembly to inflation tube.** a. Make an insertion tool from a coat hanger or any other suitable wire by clipping and filing the end round and smooth. b. Insert the insertion tool into the folded end of the spacer tube (see A-A of AF Drawing 74204). Push the spacer assembly into the bladder inflation hose, working the spacer tube in slowly so that it lies adjacent to the existing space tube spacer and so that the spacer extends 4 inches into the body bladder. When the spacer tube is completely inserted, hold on to the folded end of the spacer tube from the outside of the body bladder and pull the insertion tool out. c. See section U-U of FIGURE 2. Pull the bladder inflation tube over the spacer tube assembly and rubber sleeve flush with the wide labeled part of the connector. Clamp the bladder inflation tube to the rubber sleeve of the connector assembly with NAS 397-16 (see Operation 38) clamp toward the connector and wrap the clamped area with at least two full turns of wrapping tape. Clamped area shall be centered within at least a one inch width of wrapping tape.			

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TABLE III. Manufacturing operations. – Continued.

No.	Description of Operation	Stitch Type	Seam & Stitch Type	Stitches per Inch
38.	**Attach connector assembly to inflation tube. - contd.** d. See section U-U of FIGURE 2. Install the rubber band (see 3.4.22) over all wrapped, clamped areas so that about a third of the rubber band covers the wide labeled part of the connector and the rest covers the wrapped clamped areas. All wrapping tape shall be covered by the rubber band.			
39.	**Make connector assembly.** See section U-U of FIGURE 2. Insert the rubber sleeve (see 3.4.17), onto connector (see 3.4.16). Slide Oetiker clamp (see 3.4.18), over inflation sleeve assembly; tube covering (3.4.1.1), inflation tube (3.4.1.2), spacer (3.4.5), mesh (3.4.21) and springs (3.4.20) in preparation for assembly to connector. Connect attachment sleeve assembly to connector by inserting interlaced springs from the unfolded end of the mesh spacer tube (section B-B of Drawing 74204), into the rubber sleeve so that end of springs is about 0.25 inch from the wide labeled part of the connector. Slide spacer and mesh over connector to about 1" from the wide labeled part of the connector and wrap with at least 2 wraps of tape (see 3.4.9). Slide inflation tube to be even with the wide labeled part of the connector and wrap with at least 2 wraps of tape. Slide tube covering to be even with wide labeled part of the connector and clamp Oetiker clamp in place, positioned halfway on the connector extension and wrap with at least 2 wraps of tape. Cover tape over clamp with rubber band (see 3.4.22).	301	SSa-2	10-12

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TABLE IV. Dimensions of hook and pile fastener tape (inches).

Size of Anti-g Garment	Body Lacer Covers 1/ 2/	Thigh Lacer Covers 1/ 2/		Calf Lacer Covers 1/ 2/
Small regular (SR)	1 x 9.50	1 x 10.625		1 x 12.50
Small long (SL)	1 x 9.50	1 x 11.50		1 x 13-18
Medium regular (MR)	1 x 10	1 x 10.875		1 x 12.50
Medium long (ML)	1 x 10	1 x 11.375		1 x 13.125
Large regular (LR)	1 x 10	1 x 10.875		1 x 12.50
Large long (LL)	1 x 10	1 x 11.25		1 x 13.25
Large extra-long (LXL)	1 x 10	1 x 11.25		1 x 17.25

NOTES:

1/ Since the requirements of TABLE III specify that the fastener tape for the lacer covers shall be trimmed, if necessary, so that the fastener tape will be even with the top and the bottom edges of the flap, these dimensions are approximate.

2/ Two for each anti-g garment shall be required.

TABLE V. Length of adjustment laces (feet). 1/

Size of Anti-g Garment	Body Area 2/	Thigh Area 2/	Calf Area 2/
Small regular (SR)	12	12	14
Small long (SL)	12	13	16
Medium regular (MR)	12	13	16
Medium long (ML)	12	14	17
Large regular (LR)	13	14	17
Large long (LL)	13	16	19
Large extra-long (LXL)	13	16	19

NOTES:

1/ The dimensions specified in this table are approximate and shall be governed by the following requirement: Both ends of each adjustment lace shall extend at least six inches beyond the open end of the lacing tape when the adjustment laces are completely extended. Both ends of each adjustment lace shall be dipped in a hot solution of 50% beeswax and 50% paraffin.

2/ Two per anti-g garment shall be required.

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3.7. Performance

3.7.1. Leakage.
When the bladder in the anti-g garment is inflated to a pressure of 12 pounds per square inch gauge (psig), the bladder shall not lose more than 0.5 psig of air pressure in 60 seconds (see 4.6.2).

3.7.2. Endurance.
When the anti-g garment is inflated 1,000 times, to a pressure of 15 psig, the anti-g garment shall not develop any structural defects. Structural defects shall include, but not be limited to, material torn, seam separation, slide fastener slider lock broken, slide fastener chain separated, snap fasteners unsnapped, broken stitches, or clamps loosened or broken. After the anti-g garment has been inflated 1,000 times, to a pressure of 15 psig, the anti-g garment shall meet the requirements in Operation 14 (see 4.6.3).

3.7.3. Inflation time.
With a free airflow of 10 cubic feet per minute and with a back pressure of 9 to 12 psig applied from the pressure source, the anti-g garment shall inflate to a pressure of 8 psig in not more than 3 seconds (see 4.6.4).

3.7.4. Low temperature storage.
The anti-g garment shall not be adversely affected by storage at a temperature of −60 °F (see 4.6.5).

3.7.5. High temperature storage.
The anti-g garment shall not be adversely affected by storage at a temperature of +160 °F (see 4.6.6).

3.7.6. Low temperature operation.
The anti-g garment shall operate satisfactorily at a temperature of −30 °F (see 4.6.7).

3.7.7. Bond strength (bladder).
The force required to separate the bonded bladder halves shall be not less than 40 pounds (see 4.6.8).

3.8. Finished measurements.
The finished measurements of the anti-g garment shall be as specified in TABLE VI and shall be taken with a calibrated measuring device. The minimum finished measurements specified in TABLE VI shall be the measurements of the finished anti-g garment when its adjustment laces are tight, and the thigh take-up slide fasteners are completely closed. The maximum finished measurements specified in TABLE VI shall be the measurements of the finished anti-g garment when its adjustment laces are loose, the thigh take-up slide fasteners are completely open, and the fabric is stretched to the limit of the solid fabric with the lacer cover closed.

The finished measurements shall be taken with the body slide fastener and the leg slide fasteners open and as follows: Measure the body circumference straight across the top edge of the body from the outside edge of the zipper teeth on both sides (dimensions A and B on FIGURE 10). Measure the thigh circumference, with the thigh take-up slide fasteners closed for the minimum finished measurement and with the thigh take-up slide fastener open for the maximum finished measurement, straight across the top edge of the cut out from the outside edge of the zipper teeth

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on both sides (dimensions C and D on FIGURE 10). Measure the leg length at the slide fastener location from the top to the bottom of the slide fastener of the leg opening (dimension E on FIGURE 10). Measure the ankle circumference straight across the leg terminal from the outside of the zipper teeth on both sides (dimensions F and G on FIGURE 10). Measure the length of the inflation tube from the end of the nose piece of the connector (see FIGURE 1 of MIL-C-83390) to the end of the stitching of the bladder cover to the body front at the bladder neck opening (see body front pattern) (dimension H on FIGURE 10).

TABLE VI. Finished Measurements (inches) (see FIGURE 10).

Size of Anti-g Garment	Body Circumference		Thigh Circumference		Leg Length at Fastener Location	Ankle Circumference		Hose Length
	A (min)	B (max)	C (min)	D (max)	E (nominal)	F (min)	G (max)	H (nominal)
Small regular (SR)	29.50	34.50	21	26	26	12	15	17
Small long (SL)	29	34.25	20.75	25.50	28	12	15	17
Medium regular (MR)	32.25	38.25	22.25	27.25	26.50	12	15	17
Medium long (ML)	32	38	22.50	27.50	28.50	12	15	17
Large regular (LR)	35.25	41	23.50	28.75	27	12	15	17
Large long (LL)	35.50	41.25	24	29	29	12	15	17
Large extra-long (LXL)	35.50	41.25	24	29	33	13.75	16.75	17
Tolerance	+0.75 -0.50	+0.75 -0.50	+0.50 -0.375	+0.50 -0.375	±0.50	+0 -0.50	±0.25	±0.25

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TABLE VII. Elastic kneeboard subassemblies.

Size of anti-G Garment	Cut Length Elastic (Inches)	Cut Length Webbing (Inches)
	2 each	4 each
Small Regular	7.75	2.75
Small Long	7.75	2.75
Medium Regular	8.25	2.75
Medium Long	8.25	2.75
Large Regular	9.25	2.75
Large Long	9.25	2.75
Large Extra Long	9.75	2.75

3.9. Color.

The color of the materials of the anti-g garment shall be as specified herein. Materials for which color requirements have not been specified and which are hidden by the construction may be used in natural or colored form. The color of all exposed sewing threads shall match the color of the basic fabric specified in 3.4.1.1.

3.10. Identification of product.

3.10.1. Labels.

Each anti-g garment shall have a contractor's label, a size label, a laundering instruction label, a blank label, and an IUID label (See FIGURES 13A and 13B). All label dimensions listed refer to the final label size unless otherwise noted.

3.10.1.1. Contractor's label.

The contractor's label shall conform to MIL-DTL-32075 Type VI, Class 1 and shall be 2.50 (± 0.25) inches by 1.50 (± 0.25) inches. The label shall contain the following information:

Anti-G Garment, Cutaway, CSU-13B/P

Specification 1/

Manufacturer’s Identification 1/

Contract or Order No. 1/

1/ The manufacturer shall insert the applicable information.

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3.10.1.2. Size label.
The size label shall conform to MIL-DTL-32075 Type VI, Class 2 and shall be 2.50 (± 0.25) inches by 1.50 (± 0.25) inches. In addition to the adjective size, the size label shall contain the stature and the weight ranges (see TABLE VIII), and the national stock number (NSN) of the anti-g garment in the following manner:

Size 1/
Stature 1/ Weight 1/
NSN 1/
Lot No. 1/
Serial No. (see 3.10.1.5)
Date of Manufacture 1/
1/ The manufacturer shall insert the applicable information.

3.10.1.2.1. Stamped size label.
The upper tab of the finished bladder shall be stamped for identification with the abbreviated anti-g garment size (see TABLE VIII).

3.10.1.2.2. Stature and weight ranges.
Proper fitting of the anti-g garment is based on stature and weight; other body measurements are not used in selecting the proper size. The applicable stature and weight ranges for each size are shown in TABLE VIII.

TABLE VIII. Stature and weight ranges.

Size of anti-g Garment	Stature Range (inches)	Weight Range (pounds)
Small regular (SR)	63 – 67.875	131 – 160
Small long (SL)	68 – 72.875	131 – 160
Medium regular (MR)	64.50 – 69.375	161 – 190
Medium long (ML)	69.50 – 74.375	161 – 190
Large regular (LR)	67 – 71.375	191 – 220
Large long (LL)	71.50 – 75.375	191 – 220
Large extra-long (LXL)	75.50 - 79	191 - 230

3.10.1.3. Laundering (care) instruction label.
The laundering instruction label shall conform to MIL-DTL-32075 Type VI, Class 3 and shall be 2.50 (+ 0 or - 0.50) inches by 1.50 (± 0.25) inches. The instruction label shall contain the following information:

DO NOT IRON OR DRY CLEAN.
PLUG AIR INLET PORT SECURELY.
WASH WITH MILD SOAP AND WATER NOT OVER 120 °F.

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THE MATERIAL OF THE OUTER SHELL OF THIS
ANTI-G GARMENT IS AN INHERENT FIRE RESISTANT
MATERIAL THAT WILL NOT MELT OR DRIP AND CAN
BE LAUNDERED WITHOUT LOSING ITS FIRE-RESISTANT
PROPERTIES. NO RETREATMENT IS NECESSARY.

3.10.1.4. Blank label.
The blank label shall be 5 (± 0.25) inches by 1.50 (± 0.25) inches, conforming to MIL-DTL-32075 Type VI.

3.10.1.5. Serial numbers.
The cutaway garment shall be identified by individual serial numbers which shall be assigned by the manufacturer. Serialization shall be by a block of consecutive numbers to cover the entire acquisition document quantity.

3.10.1.6. Barcode label.
Each item shall be individually barcoded with a type VII, class 17 label in accordance with MIL-DTL-32075. The label shall be located on the bag, so it is completely visible when the item is folded or packaged as specified herein.

3.10.1.7. Item Unique Identification (IUID) label.
The contractor shall attach a scannable Item Unique Identification (IUID) label to each unit in accordance with MIL-DTL-32075, Type VI, Class 1. Per DFARS Clause 252.211-7003 the information coded onto the label shall be:

Issuing Agency Code (CAGE Code of Vendor)
Enterprise Identifier (Vendor Name)
Original Lot Number
Unique Serial Number for each item within the lot

3.10.2. Special identification.
The abdominal area of each anti-g garment that has been subjected to the endurance test (see 4.6.3) shall be marked with black or red waterproof, in letters that are at least 1 inch high. The marking shall be as follows: ENDURANCE TESTED - NOT TO BE USED IN FLIGHT.

3.11. Patterns.
The standard patterns referenced in AF Drawings 71380 through 71386 and AF Drawings 74204 and 74206 will be furnished by the Government (see 6.7). The standard patterns show size, directional lines, and notches for the proper assembly of the parts of the anti-g garment. The standard patterns shall be used by the manufacturer as a guide in cutting the manufacturer's working patterns. The manufacturer's working patterns shall be identical in size and shape to the standard patterns. Neither the standard patterns nor the manufacturer's working patterns shall be altered in any way unless approved by the engineering support activity (ESA).

3.11.1. Directional line.
A straight line with an arrow at each end and the word "straight" affixed to the straight line marked on a pattern part indicates the straight of the material which is also defined as the warp or lengthwise direction of the cloth or fabric. The pattern part shall be laid on the cloth so that the

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straight line with the arrows indicating the “straight” of the fabric is parallel to the selvage edge (lengthwise) of the fabric. This shall be accomplished by placing the straight line on the lengthwise grain of the fabric as follows: Measure the distance, at two points, from the straight line to the selvage. Adjust the pattern part until the distance from the straight line to the selvage is the same from both points (see FIGURE 11).

3.12. Workmanship.

In as much as the correct functioning of the anti-g garment directly affects the safety of personnel, it shall be free of all loose thread, lint, foreign matter, blemishes, irregularities, or defects that would adversely affect performance, appearance, or durability.

3.13. Post endurance inspection.

After completion of the endurance test, the suit shall be subjected to an additional leakage test (see 4.6.3) and the bladder post endurance inspection (see 4.6.3.1).

4. VERIFICATION

4.1. Classification of inspections.

The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3)

b. First article inspection (see 4.4)

c. Conformance inspection (see 4.5)

4.2. Inspection conditions

4.2.1. Atmospheric conditions.

Unless otherwise specified herein, all tests required by this specification, except visual and dimensional examination, shall be accomplished at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77 (± 10) °F, and at a relative humidity of 80% or less. Where tests are accomplished at atmospheric pressure or temperature different from these values, proper allowance shall be made for the change in instrument reading. Such allowances shall be approved by the ESA.

4.2.2. Connector.

A connector conforming to MS27755 shall be used to connect the anti-g garment to the air source for tests that require inflation and deflation cycling of the anti-g garment.

4.2.3. Air.

The air used to test the anti-g garment as specified herein shall not contain any oil or moisture.

4.2.4. Air pressure gauge.

Air pressure gauges utilized shall be in pounds per square inch (psig) and shall be precise and calibrated to at least one hundredth of a psig. The gauge shall have a current calibration.

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4.3. Qualification inspection.

The qualification samples described in 4.3.1 shall be subjected to qualification inspection. The qualification inspection for the anti-g garments and bladders shall consist of the test methods described in 4.6.1 through 4.6.7. The qualification inspection for the bladder shall consist of the test described in 4.6.8 (see TABLE IX). The garments for inflation time and endurance testing shall be fit to inanimate models in accordance with 4.7, 4.7.1, and TABLE IX.

4.3.1. Qualification samples.

Qualification samples, consisting of two anti-g garments tested as specified in 4.5 and of two untested bladders, shall be furnished to the qualifying activity (see 6.3).

TABLE IX. Qualification inspections.

Inspection	Requirement	Test Method
Visual examination	3.5	4.6.1 & TABLE VIII
Leakage	3.7.1	4.6.2
Endurance	3.7.2	4.6.3
Inflation time	3.7.3	4.6.4
Low temperature storage	3.7.4	4.6.5
High temperature storage	3.7.5	4.6.6
Low temperature operation	3.7.6	4.6.7
Bond strength (bladder)	3.7.7	4.6.8
Finished measurements	3.8 & TABLE VI	4.7 & FIGURE 10
Bladder post-endurance	3.13	4.6.3.1

4.4. First article inspection.

The first article samples specified in 4.4.1.1 shall be subjected to first article inspection. The first article inspection for the anti-g garment shall consist of the tests described in 4.6.1 through 4.6.7. The first article inspection for the bladder shall consist of the test described in 4.6.8 (see TABLE X). The garments for inflation time and endurance testing shall be fit to inanimate models in accordance with 4.7, 4.7.1, and TABLE XV.

4.4.1. First article samples.

The first article samples shall be manufactured after the award of the contract and before production of the anti-g garment commences. The first article samples shall consist of the items specified in 4.4.1.1 and shall be representative of the materials, components, construction, and workmanship to be used in the production items. Approval of the first article samples or the waiver of the first article inspection shall not preclude the requirements for the performance of the quality conformance inspection.

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4.4.1.1. Test samples.

The manufacturer of the anti-g garments shall subject one complete anti-g garment and one bladder for each size being made under that contract (see 6.2b) for the first article inspection described in 4.4.

TABLE X. First article inspections.

Inspection	Requirement	Test Method
Visual examination	3.5	4.6.1 & TABLE VIII
Leakage	3.7.1	4.6.2
Endurance	3.7.2	4.6.3
Inflation time	3.7.3	4.6.4
Bond strength (bladder)	3.7.7	4.6.8
Finished measurements	3.8 & TABLE VI	4.7 & FIGURE 10
Bladder post-endurance	3.13	4.6.3.1

4.5. Conformance inspection.

4.5.1. Sampling inspection.

Sampling inspection shall be conducted in accordance with plans A, B, and C for the end items and components (see 4.5.2.2, 4.5.2.3, and 4.5.2.4). Sampling inspection test requirements are described under 4.6 (see TABLE XI). At any time, the Government may choose more samples for inspection, or they may move from a normal inspection to a tightened inspection as outlined in ANSI Z1.4.

TABLE XI. Conformance inspections.

Inspection	Requirement	Test Method	Plan (ref 4.5.2.X)
Visual examination	3.5	4.6.1	C
Finished Measurements	3.8	TABLE VIII & FIGURE 10	C
Leakage	3.7.1	4.6.2	C
Endurance	3.7.2	4.6.3 & 4.6.3.1	A
Inflation time	3.7.3	4.6.4	A
Bond strength (bladder)	3.7.7	4.6.8	A
Bladder post-endurance	3.13	4.6.3.1	A
Bladder visual	FIGURE 3	FIGURE 3	B

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4.5.2. Plans for end items and components.

4.5.2.1. Lot formation.

A lot of finished anti-g garments shall consist of all suits that will be presented to the Government, including the suit(s) to be endurance tested (a lot of 150 suits would include 1 suit to be endurance tested and 149 suits to be part of the accepted/rejected lot). Each lot of completed anti-g garments shall consist of no less than 51 and no more than 500 anti-g garments that have been manufactured under the same conditions and submitted for inspection at the same time. Each lot of completed bladders shall consist of no more than 501 bladders that have been manufactured under the same conditions and submitted for inspection at the same time. Lot sizes below 500 garments and 501 bladders are acceptable. All tests outlined in plans A, B, and C shall be completed with Government representation.

4.5.2.2. Plan A.

One bladder shall be selected at random from each lot of bladders and subjected to the bond strength test described in 4.6.8.

One anti-g garment of any size manufactured in the lot shall be selected at random from each lot containing 1 to 150 anti-g garments. Two anti-g garments of any size manufactured in the lot shall be selected at random from each lot containing 151 to 500 anti-g garments. Each anti-g garment selected shall be subjected to the following tests as described under 4.6:

a. Endurance (4.6.3)

b. Bladder post endurance inspection (4.6.3.1)

c. Inflation time (4.6.4).

4.5.2.3. Plan B.

From a lot of bladders, a sample shall be chosen for inspection in accordance with ASQ Z1.4. The bladders shall be inspected with a special inspection level of S-3 and shall be subjected to a visual examination to ensure compliance with FIGURE 3. If any non-compliant bladders are discovered in the sample inspection, the lot of bladders shall be reinspected and all non-compliant bladders removed.

4.5.2.4. Plan C.

For each lot of complete anti-g garments, samples shall be chosen according to ASQ Z1.4. Samples for the visual examination of anti-g garments shall be chosen using general inspection level II and the sample size shall not be increased according to TABLE II of ASQ Z1.4 unless otherwise directed by a Government representative. For the critical characteristics listed in TABLE XIV as critical defects, use an Acceptance Quality Limit (AQL) of 0.4. For characteristics listed in TABLE XIV as major defects, use an AQL of 1.0. For characteristics listed in TABLE XIV as minor defects, use an AQL of 10. The visual samples shall be subjected to the examination described in 4.6.1.

For example, for a lot of 150 suits at general inspection level II, 20 suits shall be chosen for visual inspection. For critical defects with an AQL of 0.4, accept on 0 defects and reject on 1. For major defects with an AQL of 1.0, accept on 0 defects and reject on 1. For minor defects with an AQL of 10, accept on 5 defects and reject on 6. As another example, for a lot 500 suits at general inspection level II, 50 suits shall be chosen for visual examination. For critical defects with an

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AQL of 0.4, accept on 0 defects and reject on 1. For major defects with an AQL of 1.0, accept on 1 defect and reject on 2. For minor defects with an AQL of 10, accept on 10 defects and reject on 11.

Dimensional and leakage inspection samples shall be chosen using general inspection level I. These are critical characteristics and shall be inspected with an AQL of 0.4. The dimensional and leakage samples shall be subjected to the tests described in TABLE VII and 4.6.2.

4.5.2.5. Rejection and retest.

When one or more items selected from the submitted lot fail to meet the specification requirements for finished measurement, leakage (pre and post endurance), endurance, and inflation time, the lot shall be rejected. Visual defects classification shall be determined in accordance with paragraph 4.5.2.4 and ASQ Z1.4. If the lot is rejected, acceptance of all items in the lot shall be withheld until the extent and cause of the failure have been determined. The contractor shall explain fully to the Government representative and notify the procuring activity in writing, the cause of failure, the action taken to preclude recurrence, and the impact this failure may have in scheduled deliveries. After correction, all the sampling tests shall be repeated.

4.5.2.6. Defects in items already accepted.

The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6. Test methods.

4.6.1. Examination of product.

A sample of anti-g garments and bladders shall be examined to determine compliance with the requirements of this specification with respect to materials, finished measurements, identification marking, and workmanship. Defects found during this examination shall be classified as specified in TABLE XIV. Zero (0) critical defects and one (1) major defect are acceptable per lot of suits regardless of lot size. For a lot of 1-150 suits, four (4) minor defects per lot are acceptable. For a lot of 151-500 suits, seven (7) minor defects per lot are acceptable. Any defect beyond the acceptable level shall be cause for rejection of the entire lot.

TABLE XII. Samples needed per lot size.

Lot Size	Visual	Dimensional and Leakage	Endurance and Inflation Time
	(Gen II)	(S-4)	
1-50	8	5	1
51-90	13	5	1
91-150	20	8	1
151-280	32	13	2
281-500	32	13	2

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TABLE XIII. Lot acceptance and rejection.

	Critical		Major		Minor	
Lot Size	Ac	Re	Ac	Re	Ac	Re
1-50	0	1	1	2	4	5
51-90	0	1	1	2	4	5
91-150	0	1	1	2	4	5
151-280	0	1	1	2	7	8
281-500	0	1	1	2	7	8

TABLE XIV. Classification of defects.

NO.	DEFECT 1/	CRITICAL	MAJOR	MINOR
1.	**Materials**			
	a. Materials not conforming to specified requirements unless otherwise classified herein.	1		
2.	**Fabric imperfection**			
	b. See FED-STD-4 2/		101	
	- On lacer cover			201
	- All other areas			
3.	**Slide fasteners 3/**			
	a. Fabric seam on tape catching in slider or slider not operating freely.		102	
	b. Slide fastener not specified type, style, or length.	2		
	c. Any part of slide fastener assembly omitted, bent, or broken.	3		
	d. Separation of slide fastener chain	4		
	e. Pull tab missing.		103	
	f. Stiffener missing.	5		
	g. Pull tab not specified type.		104	
	h. Stiffener not specified type.		105	
4.	**Color**			
	a. Base material not specified color.	6		
	b. Any component color not approximately matching specified color shade.		106	

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TABLE XIV. Classification of defects. – Continued.

5.	**Snap fasteners 4/**			
	a. Stud not securely clenched, damaged spring, or damaged or missing snap fastener.		107	
	b. Stud not aligned with the socket causing noticeable bulge or twist when snapped.		108	
	c. Stud mismatched with socket.		109	
	d. Snap fastener not holding together.		110	
	e. Assembled snaps rotate when hand force is applied for both NASM27983 and NASM27980.		111	
	f. Snap fastener not specified type.	7		
	g. Split and/or crack in barrel as noticed by individual.	8		
	h. Snap fasteners not separating under normal pull.		112	
	i. The roll crimp is not a continuous sloped circle.		113	
6.	**Assembly**			
	a. Incorrectly assembled, that is, reversed seams, not specified type of seam, not specified type of stitch, or misplaced or reversed components, etc.	9		
	b. Incompletely assembled, that is, any component or required operation omitted.	10		
7.	**Identification of product**			
	a. Any label missing, misplaced, or illegible.		114	
	b. IUID label will not scan.		115	
	c. Information on any label incorrect or incomplete.		116	
8.	**Workmanship**			
	a. Hole, cut, tear, or needle chew.	11		
	b. Darn or mend affecting serviceability.	12		
	c. Darn or mend affecting appearance.		117	
	d. Stain, oil spot, or dirt over 0.25 inch diameter (per instance).		118	
	e. Stain, oil spot, or dirt up to or equal to 0.25 inch diameter in not more than one place on suit.			202

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TABLE XIV. Classification of defects. – Continued.

NO.	DEFECT 1/	CRITICAL	MAJOR	MINOR
8.	**Workmanship – contd.**			
	f. Stain, oil spot, or dirt up to or equal to 0.25 inch in diameter in two or more places on one suit.		119	
	g. Thread end not trimmed to a maximum of 0.25 inch (one, two, or three instances on one suit).			203
	h. Thread end not trimmed to a maximum of 0.25 inch (more than three instances on one suit).		120	
	i. Twisting or distortion of the garment during testing.	13		
	j. All other areas		121	204
	NOTE: These defects shall be classified and scored as a major defect when it affects the serviceability; it shall be classified and scored as a minor defect when it affects appearance.			
9.	**Measurements**			
	a. Any finished measurement not within specified tolerance.	14		
	b. Distance between inside longitudinal stitch lines, attaching knife pocket to reinforcement patch, greater than 2.375 inches or less than 2.125 inches.		122	
	c. Width of inside opening of knife pocket (distance between longitudinal stitch lines) less than 1.75 inches.		123	
10.	**Hook assembly 5/**			
	a. Hook and loop assembly not specified type.	15		
	b. Hook not aligned with loop (tolerance 0.125 inch)			
	- More than 0.125 inch but less than 0.25 inch.			205
	- Equal to or more than 0.25 inch.		124	
	- Affecting operation.	16		
	c. Visual defects			
	- Affecting appearance.			206
	- Affecting serviceability.		125	

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TABLE XIV. Classification of defects. – Continued.

NO.	DEFECT 1/	CRITICAL	MAJOR	MINOR
11.	**Stitches and seams**			
	a. Stitching missing or incomplete.	17		
	b. Stitching broken or cut.			
	- On lacer cover, slide fastener pull tape, slide fastener loops			207
	- On snap tabs, labels		126	
	- All other areas	18		
	c. Skipped stitch			
	- On lacer cover, slide fastener pull tape, slide fastener loops			208
	- On snap tabs, labels		127	
	- All other areas	19		
	d. Run-off.	20		
	e. Open or separated seam.	21		
	f. End of stitching not securely backstitched as specified.		128	
	g. Operations out of tolerance			
	- Less than two times the allowable tolerance on the positive or negative side.			209
	- Two to three times the allowable tolerance on the positive or negative side.		129	
	- Over three times the allowable tolerance on the positive or negative side.	22		
	- Any stitching or seam less than 0.031 inch from the edge for more than one consecutive stitch measuring less than or equal to an inch in length.		130	
	- Any stitching or seam less than 0.031 inch from the edge measuring over an inch in length.	23		
	h. More than 0.25 inch but less than or equal to 1 inch raw edge measured in length or width.			210
	i. More than 1 inch raw edge measured in length or width.		131	

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TABLE XIV. Classification of defects. – Continued.

NO.	DEFECT 1/	CRITICAL	MAJOR	MINOR
12.	**Bladder**			
	a. Spacer material or the attachment patches caught in any portion of the welded seam	24		
	b. The seams of the inflated bladder not properly joining adhering. The seams not straight, continuous, and parallel to the edge of the bladder.	25		
	c. The distance from the outside edge of the weld to the edge of the bladder material greater than 0.50 inch or less than 0.063 inch		132	
	d. The upper tab of the finished bladder shall be stamped.		133	
	e. Bladder fin edge (not tab) caught in body bladder cover stitching.	26		
	f. Puckering of the bladder or material at the bladder tabs.		134	
13.	**Leather**			
	a. Hole in leather (needle hole in less than three places) other than perimeter stitching.			211
	b. Hole in leather (needle hole in three or more places) other than perimeter stitching.		135	
	c. Hole in leather (hole larger than a needle).	27		
	d. Scuff on leather (less than one square inch per instance).			212
	e. Scuff on leather (greater than one square inch per instance).		136	

NOTES:

1/ All defects are counted per instance unless otherwise noted.

2/ Fabric imperfections that would affect the function of the anti-g garment shall be considered major defects.

3/ The slide fastener shall be checked for proper function and attachment by zipping closed and unzipping the slide fastener at least two times.

4/ The snap fastener shall be checked for proper function and attachment by snapping closed and unsnapping the snap fastener at least three times.

3/ The hook assembly shall be checked by attaching the hook to the loop and closing the slide fastener. Alignment of the hook assembly shall not interfere with the slide fastener operation.

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4.6.2. Leakage.
The bladder in the anti-g garment shall be inflated to a pressure of 12 psig by means of a suitable connection arranged so that the pressure of the bladder will be indicated by a gage. The air supply shall be shut off securely. After a minimum of 60 seconds the pressure in the bladder shall be checked and readjusted, if necessary, to the original pressure of 12 psig. A minimum of 60 seconds after the pressure has been checked and readjusted, the pressure in the bladder shall be checked. If the pressure in the bladder is less than 11.5 psig, the anti-g garment shall be rejected. While the anti-g garment is still inflated, it shall be examined for material and constructional failure, separation of seams and components, change in location of slide fastener sliders, separation of slide fastener chains, and twisting and distortion of the anti-g garment. If the anti-g garment contains any of these defects, it shall be rejected. After this examination is completed the bladder of the anti-g garment shall be completely deflated.

4.6.3. Endurance.
The anti-g garment shall be fitted to the inanimate model specified in 4.7. The anti-g garment shall be inflated 1,000 times, in accordance with TABLE XV, to a pressure of not less than 15 psig and then examined for structural defects. If the anti-g garment contains any of the defects specified in 4.6.1, the anti-g garment shall be rejected. After the anti-g garment has been examined for defects, the anti-g garment shall be subjected to the leakage test described in 4.6.2. Any anti-g garment that has experienced more than 100 inflation test cycles at 11 psi or greater shall be deemed as having undergone destructive testing. This garment is not acceptable for shipment against the quantities of serviceable finished end items required by contract and shall be marked in accordance with paragraph 3.10.2 as having undergone destructive testing.

TABLE XV. Inflation of anti-g garment.

Minimum pressure (psig)	Maximum seconds permitted to inflate garment to minimum pressure	Minimum seconds permitted to deflate garment to zero psig (gage pressure reading)	Minimum cycles
15	8	4	5 per minute
12	8	4	5 per minute
6	30	10	10 per hour

4.6.3.1. Bladder post endurance inspection.
After the successful completion of endurance testing and successful completion of post endurance leakage, the bladder assembly shall be carefully removed from the casing assembly of the endurance tested anti g garments. To remove the bladder from the anti g casing carefully remove the stitching on the (9) bladder tabs. Remove the rubber sleeve and clamps or clamps and tape from the hose, separate the connector from the hose and remove the spring. Make a 5 inch incision in the casing material only, on the inside of the abdomen bladder. NOTE: Use extreme care not to cut the bladder or any stitching. Remove the complete bladder through this incision. The bladders shall be dimensionally and visually inspected in accordance with TABLES III and X and FIGURES 3 and 4.

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4.6.4. Inflation time.
The anti-g garment shall be fitted to the inanimate model specified in 4.7. The anti-g garment shall be connected to a source of air with a free flow of 10 cubic feet per minute. The regulator shall be adjusted so that the pressure shall not exceed 12 psig. The bladder in the anti-g garment shall inflate to a pressure of 8 psig in not more than 3 seconds when the pressure is applied.

4.6.5. Low temperature storage.
The anti-g garment shall be packaged in a corrugated box with maximum dimensions of 13.50 by 4.50 by 8.50 inches. The packaged anti-g garment shall be subjected to a temperature of -60 °F or colder for at least 48 hours. The packaged anti-g garment shall then be kept at room temperature for at least 24 hours. The anti-g garment shall be removed from the box, fitted to the inanimate model specified in 4.7 and inflated 500 times, in accordance with TABLE XV, to a pressure of not less than 12 psig. At the completion of this test, the anti-g garment shall be examined for failures or defects in the material or workmanship. Structural defects or any other defects shall be cause for rejection of the anti-g garment.

4.6.6. High temperature storage.
The anti-g garment shall be packaged in a corrugated box with maximum dimensions of 13.50 by 4.50 by 8.50 inches. The packaged anti-g garment shall be subjected to a temperature of +160 °F or warmer for at least four (4) hours. The packaged anti-g garment shall then be kept at room temperature for at least 24 hours. The anti-g garment shall be removed from the box, fitted to the inanimate model specified in 4.7 and inflated 500 times in accordance with TABLE XV to a pressure of not less than 12 psig. At the completion of this test, the anti-g garment shall be examined for failures or defects in the material or workmanship. Structural defects or any other defects shall be cause for rejection of the anti-g garment.

4.6.7. Low temperature operation.
The anti-g garment shall be subjected to a temperature of -30 °F or colder for a period of four (4) hours. During the four-hour period the anti-g garment is being subjected to the -30 °F, it shall be inflated at the rate of 10 times per hour to a pressure of not less than six (6) psig in accordance with TABLE XV. At the completion of this test, the anti-g garment shall be examined for failures or defects in the material or workmanship. Structural defects or any other defects shall be cause for rejection of the anti-g garment.

4.6.8. Bond strength (bladder).
Five (5) specimens shall be cut from different locations of the bladder so that the specimens will be representative of the sealing of the entire bladder. The specimens shall be taken from the straight of the bladder only. Each of the specimens shall be cut across and perpendicular to the seam of the bladder and shall be 1 ±0.063 inch wide and 6 ±0.063 inches long. A suitable inspection apparatus equipped with calibrated recording mechanism (see ASTM D5034) shall be used to conduct the bond strength test. The front clamp of each jaw of the inspection apparatus shall be 1 inch by 3 inches. The rate of separation of the jaws of the inspection apparatus under no load shall be 12 ±0.50 inches per minute. One of the free ends of a specimen (an end that is not bonded) shall be placed in the upper jaw of the inspection apparatus. The other free end of the specimen shall be placed in the lower jaw of the inspection apparatus. The bonded portion (seam of bladder) of the specimen shall be in the center between the two jaws. The jaws shall be separated until either breakage of the cloth or separation of the bond occurs. The highest value required to separate the

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bonded specimens or to break the cloth shall be recorded as the test result of the individual bonded specimen. This test shall be repeated until the other four specimens have been tested for bond strength. The lowest value obtained from the tests of the five specimens shall be reported as the test result for the bond strength test and shall be not less than 40 pounds.

4.7. Inanimate models.

Two inanimate models, each consisting of two legs and one torso, shall be required. The anti-g garments in sizes small regular and small long shall be fitted to the small model. The anti-g garments in sizes medium regular, medium long, large regular, large, long, and large extra-long shall be fitted to the large model. The inanimate models shall conform to FIGURE 12 and TABLE XVI. The inanimate models shall have smooth surfaces that are free of nicks, burrs, or foreign material.

4.7.1. Fitting anti-g garment to models.

Fitting of the anti-g garments for all tests that require the use of the inanimate models shall be accomplished as follows: Wooden or plastic spacers shall be inserted between the anti-g garment and the model. The spacer for the torso section shall be 4 inches wide by 12 to 14 inches long by 0.75 inch thick and shall be inserted so that the 4 inch width will face the model. The spacer for each leg shall be 2 inches wide by 12 to 14 inches long by 0.75 inch thick and shall be inserted so that the 2 inch width will face the model. The adjustment laces of the anti-g garment shall then be taken up and tied so that the anti-g garment will fit snugly, but not tightly, around the abdomen and thighs. All slide fasteners shall be in the fully closed position. The model shall be placed in a sitting position, approximating that of a pilot seated in the cockpit of an aircraft, with a seat back angle of 15 degrees. If a snug fit cannot be obtained at the body for the sizes large long and large extra-long anti-g garments, shims shall be affixed to the model on both sides of the H and J dimensions shown on FIGURE 12. All spacers shall be removed prior to commencement of the tests.

TABLE XVI. Inanimate models dimensions.

Leg Dimensions (inches) 1/							
MODELS	**A**	**B** 2/	**C** 2/	**D**	**E**		
Small	33	6.75	2.688	1.25	1.25		
Large	33	7.188	3.063	1.25	1.50		
Torso Dimensions (inches) 1/							
MODELS	**G**	**H**	**I**	**J**	**K**	**L**	**M**
Small	14	12.75	8.063	10.125	6.813	1.25	1.25
Large	14	13.75	9.125	11.375	7.938	1.25	1.50
NOTES:							
1/ Tolerances on all dimensions shall be ±0.063 inch Except dimension G which shall have a tolerance of ± 0.25 inch.							
2/ Dimensions B and C are diameters.							

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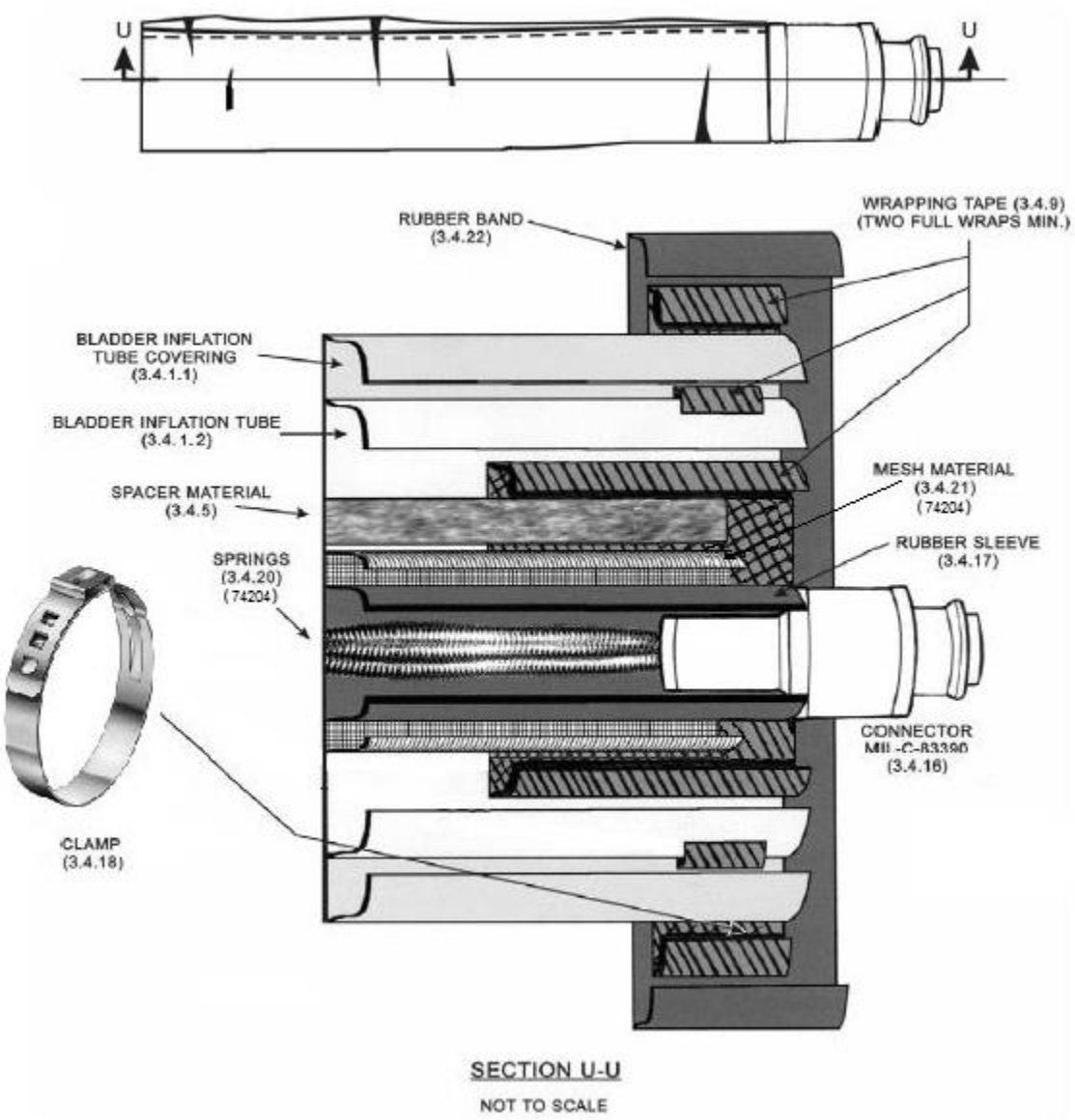


FIGURE 2. Inflation tube assembly.

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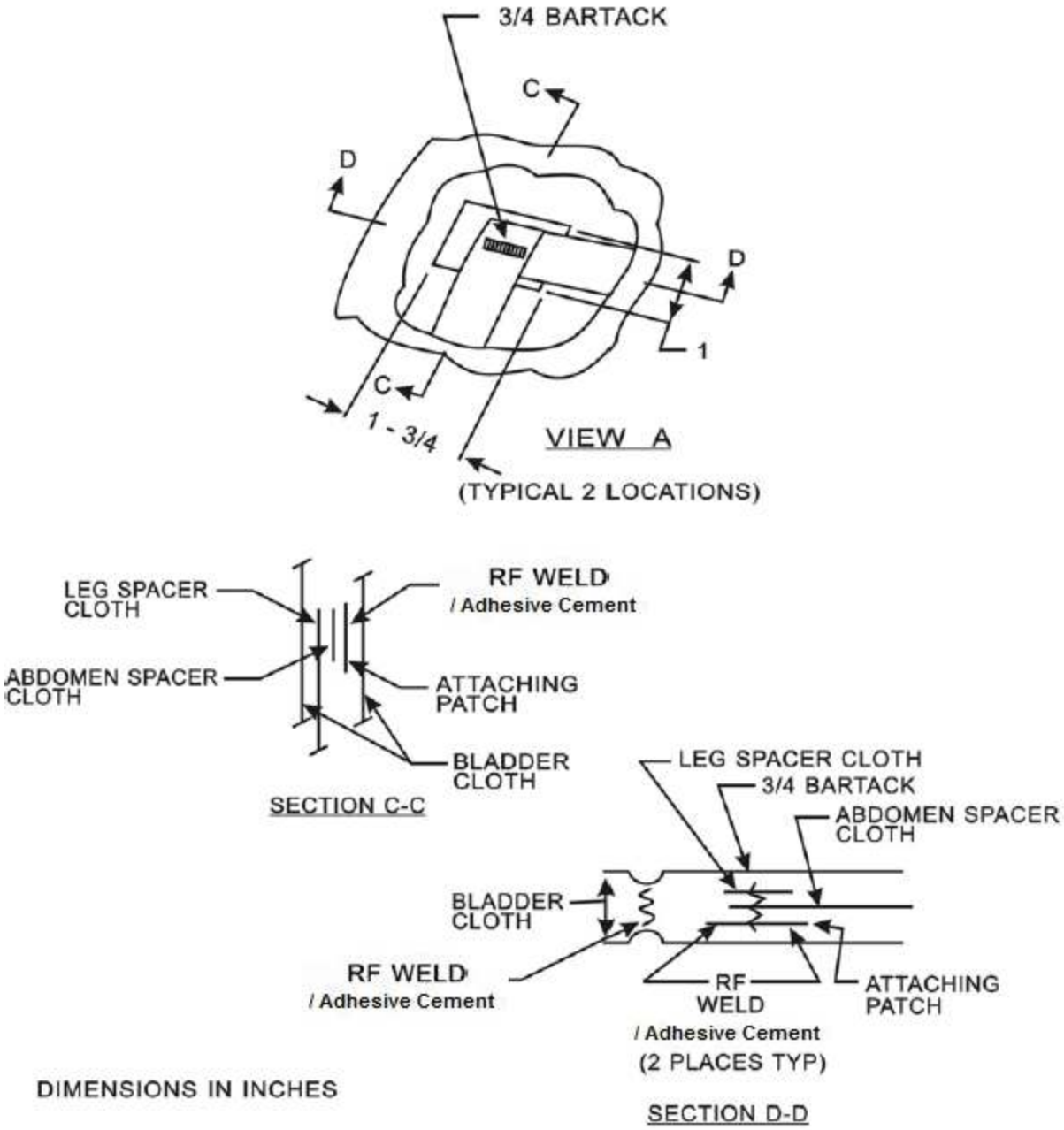


FIGURE 3. Bartack (view A).

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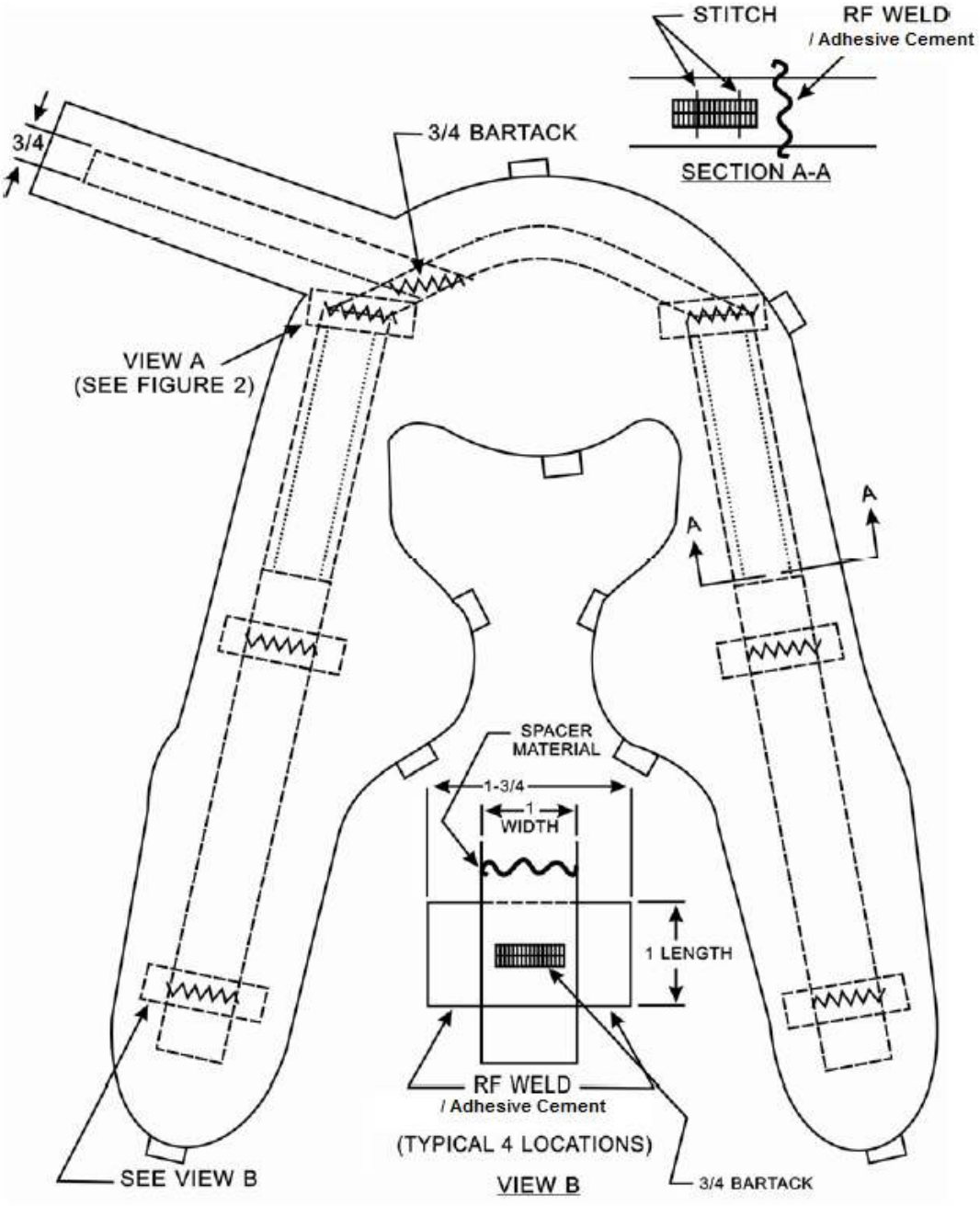


FIGURE 4. Bladder construction (dimensions are in inches).

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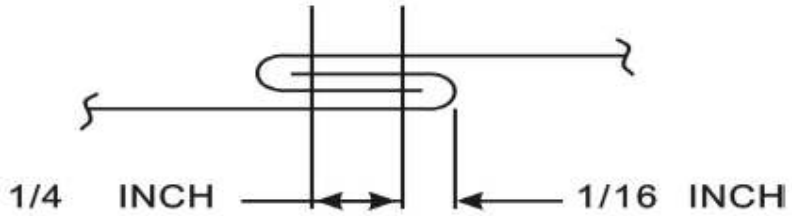


FIGURE 5. Seam type joining body front to body back.

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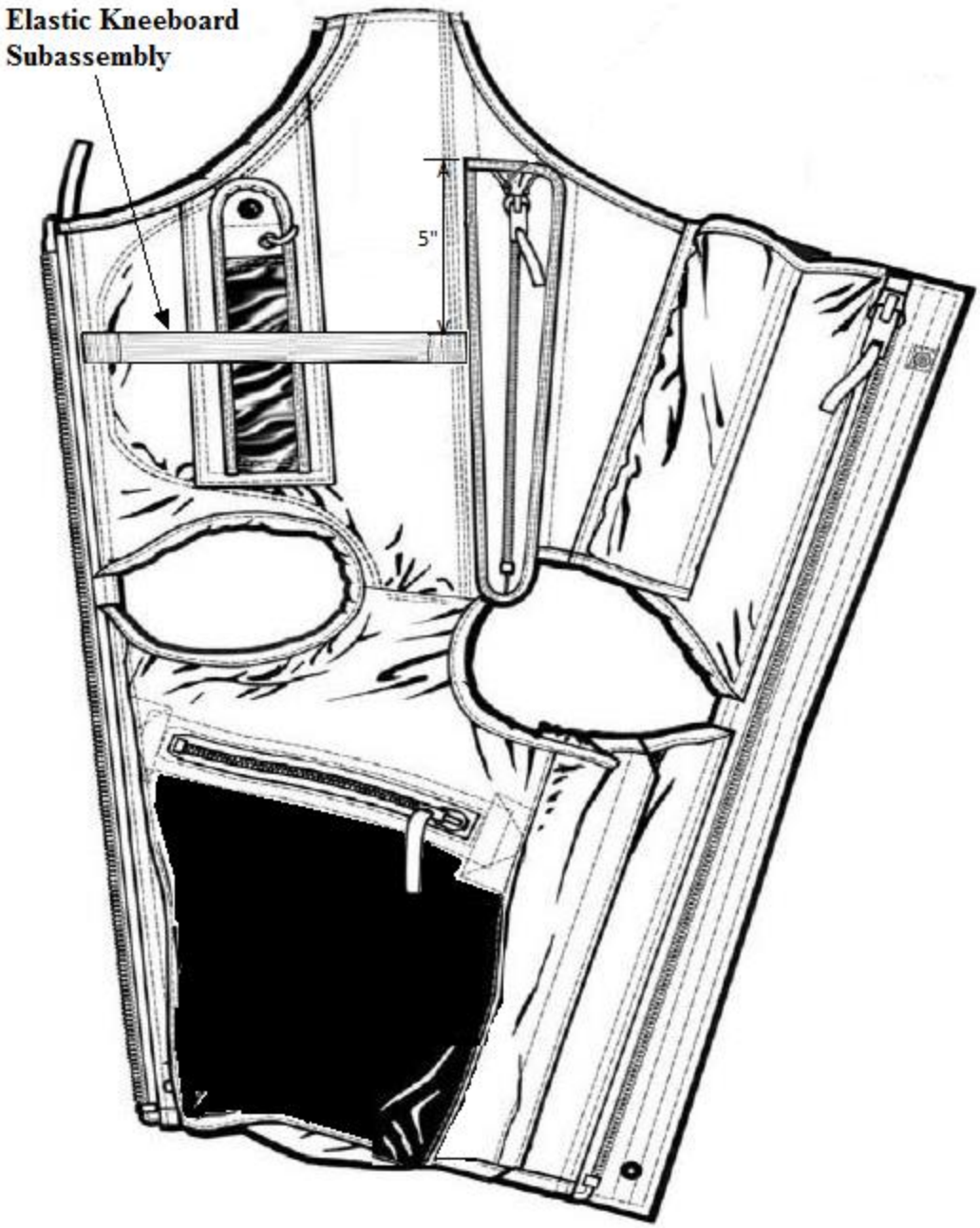


FIGURE 6. Location of kneeboard subassembly.

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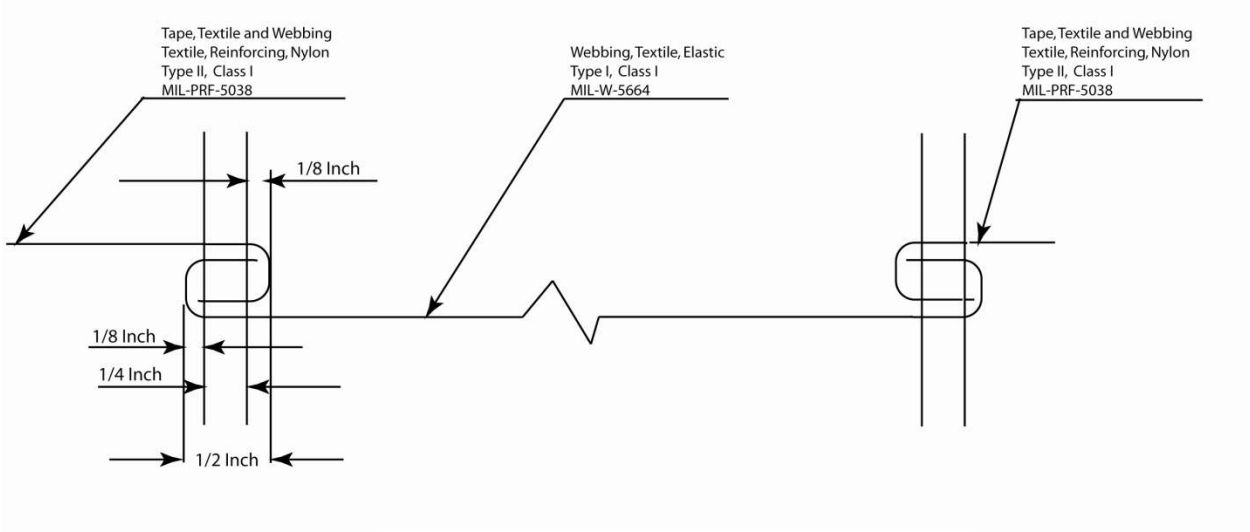


FIGURE 7. Construction of kneeboard assembly

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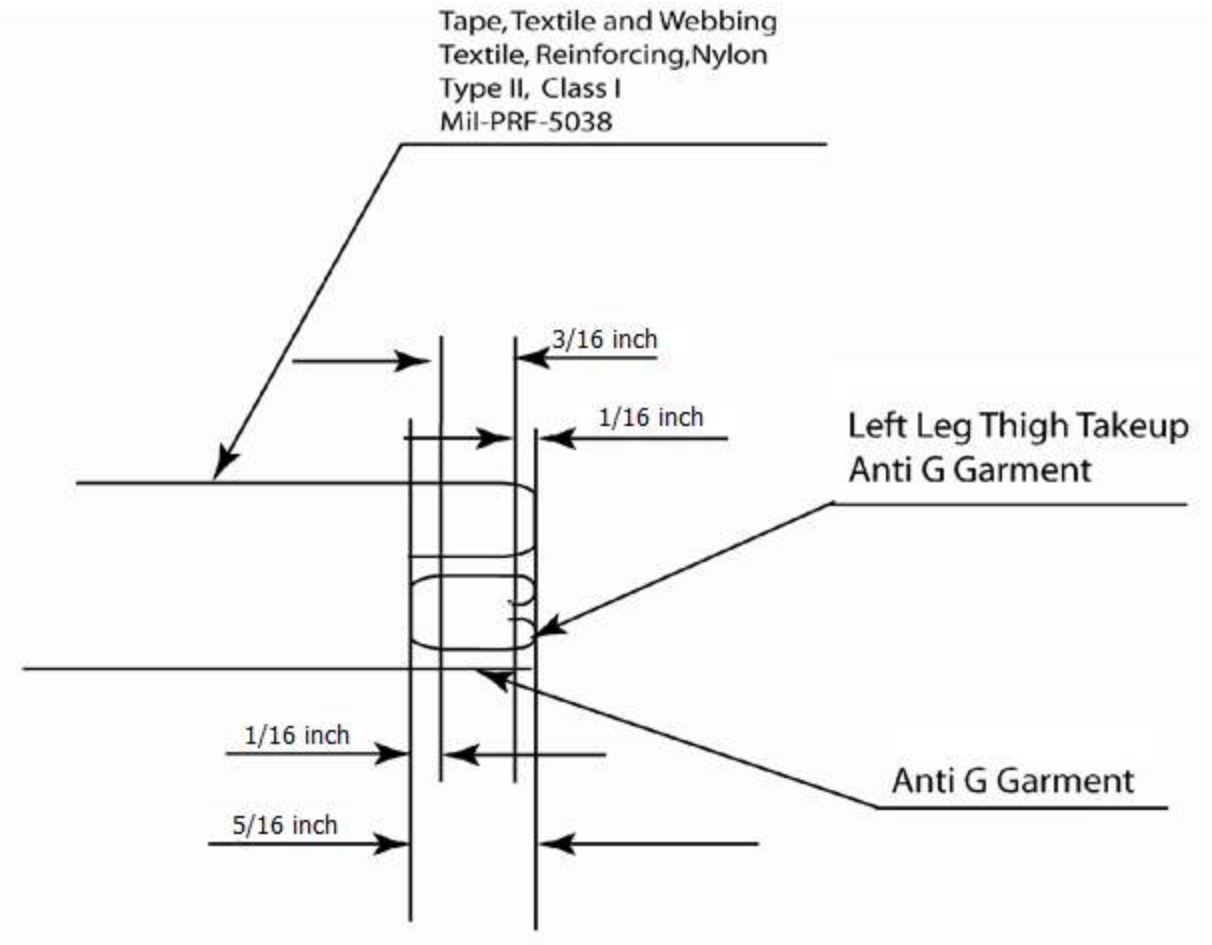


FIGURE 8. Attachment of kneeboard assembly to leg thigh take-up side.

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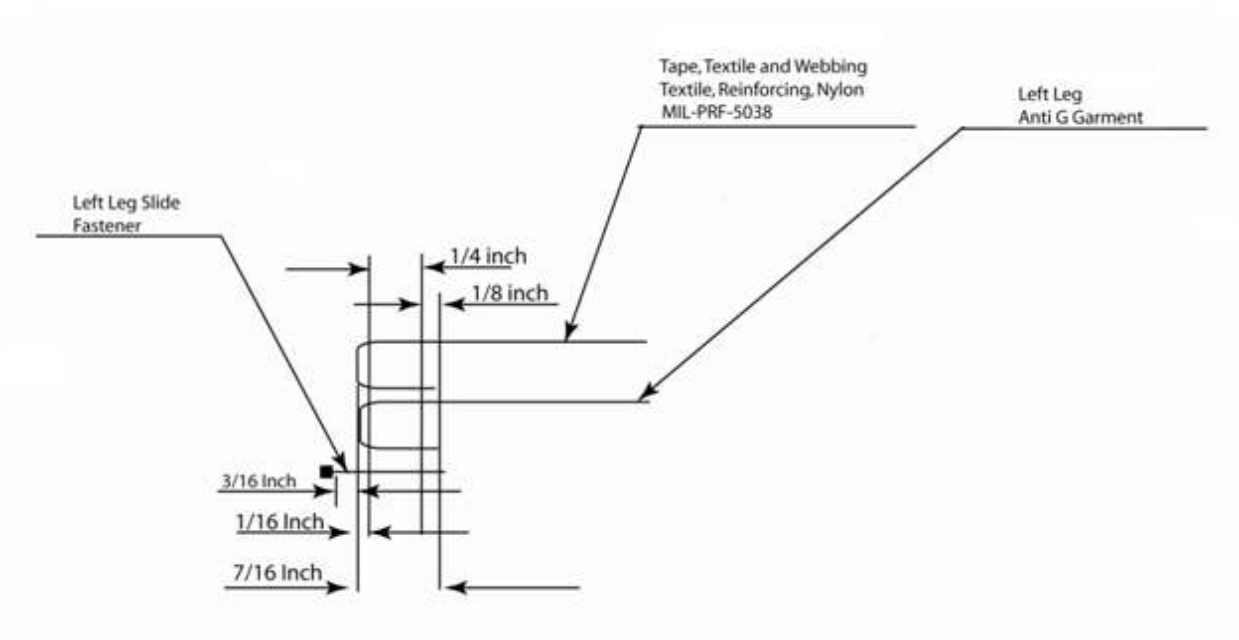


FIGURE 9. Attachment of kneeboard assembly to the leg slide fastener side.

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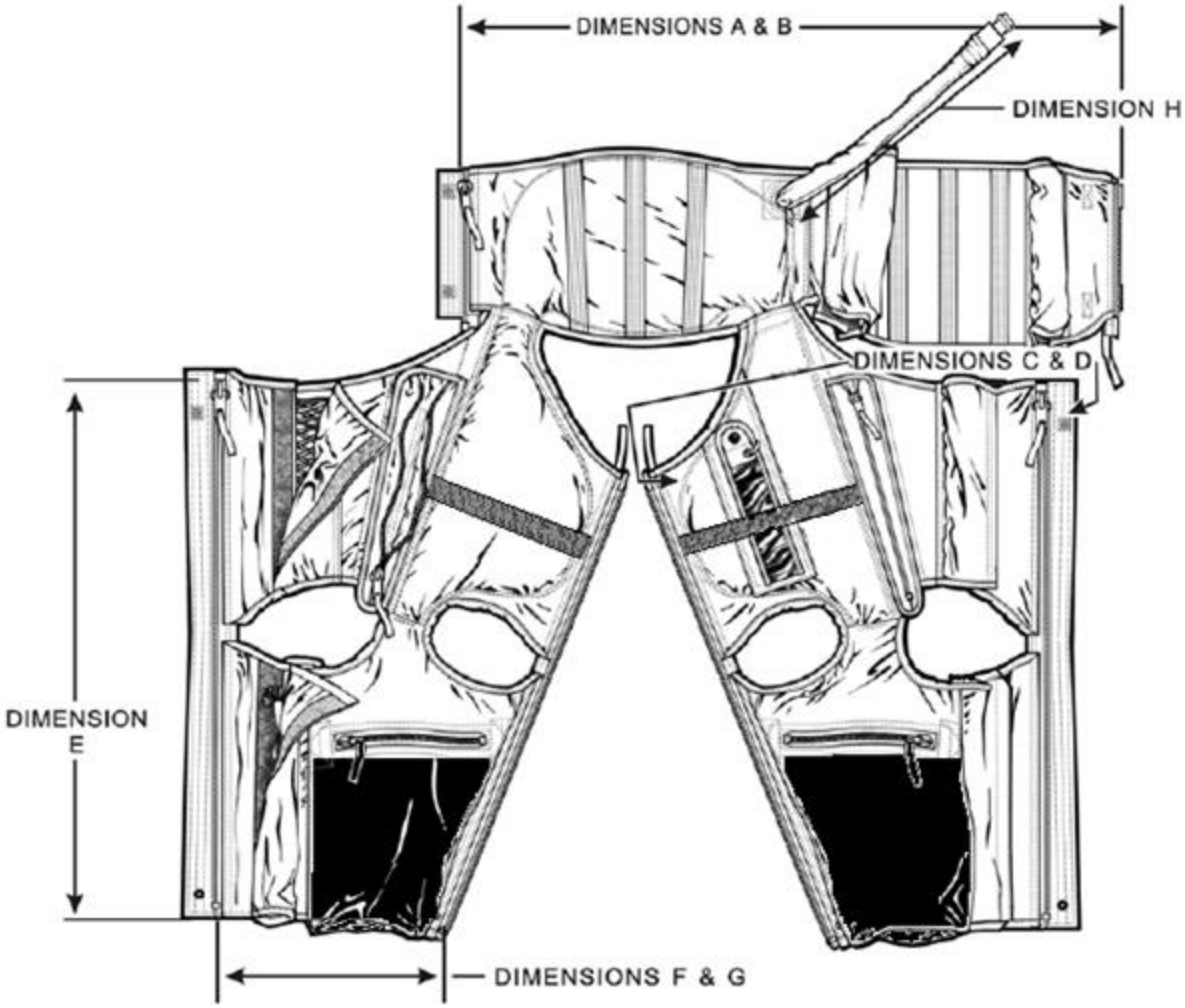


FIGURE 10. Finished garment measurements, typical (see 3.8).

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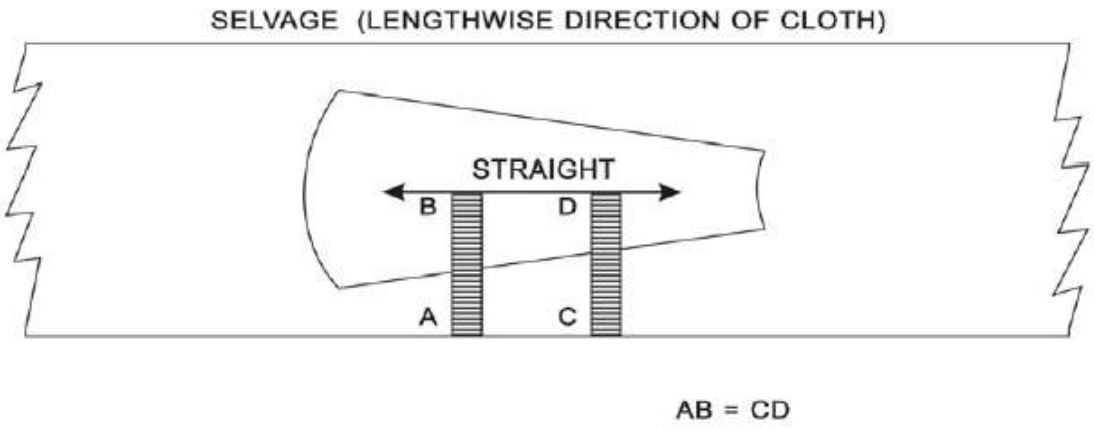


FIGURE 11. Directional line.

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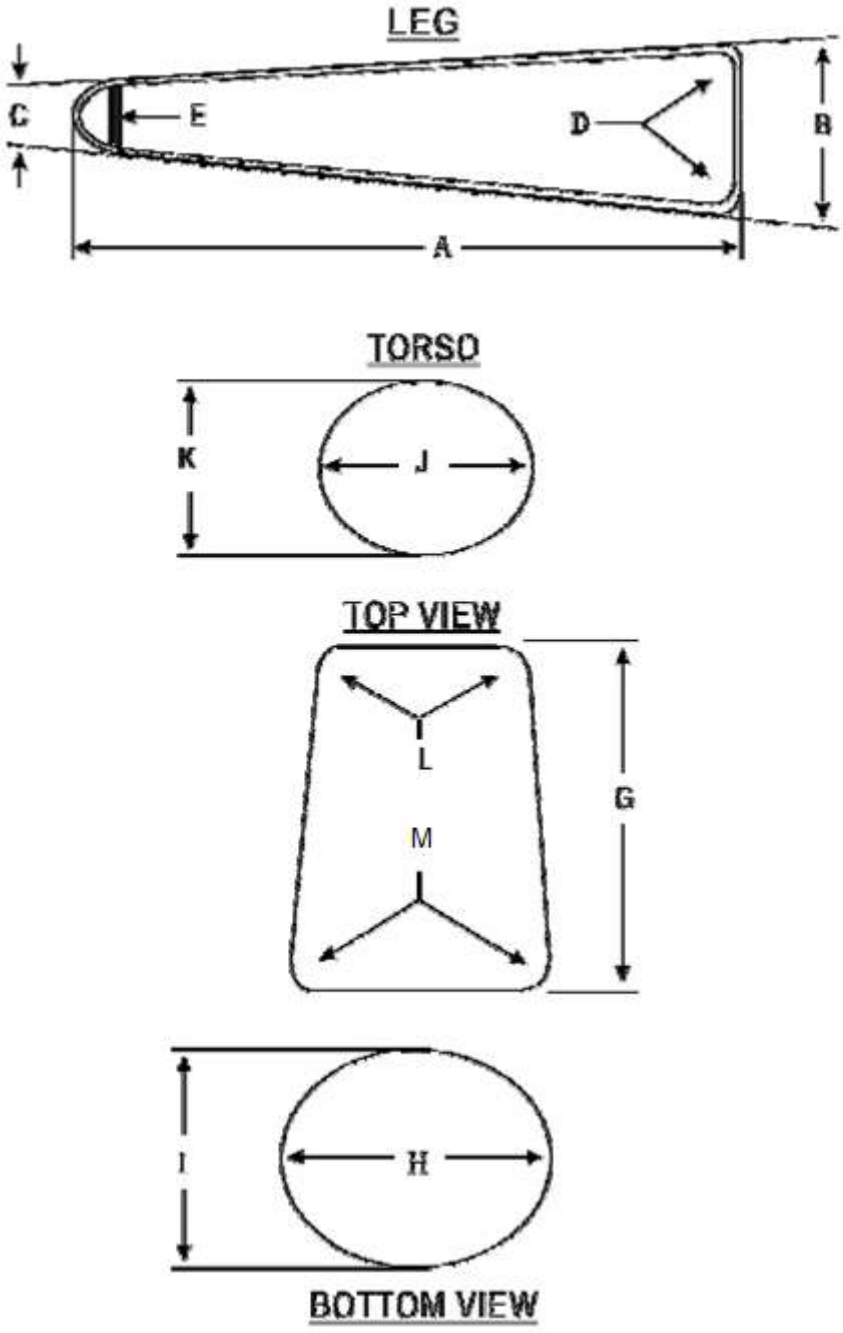


FIGURE 12. Inanimate models.

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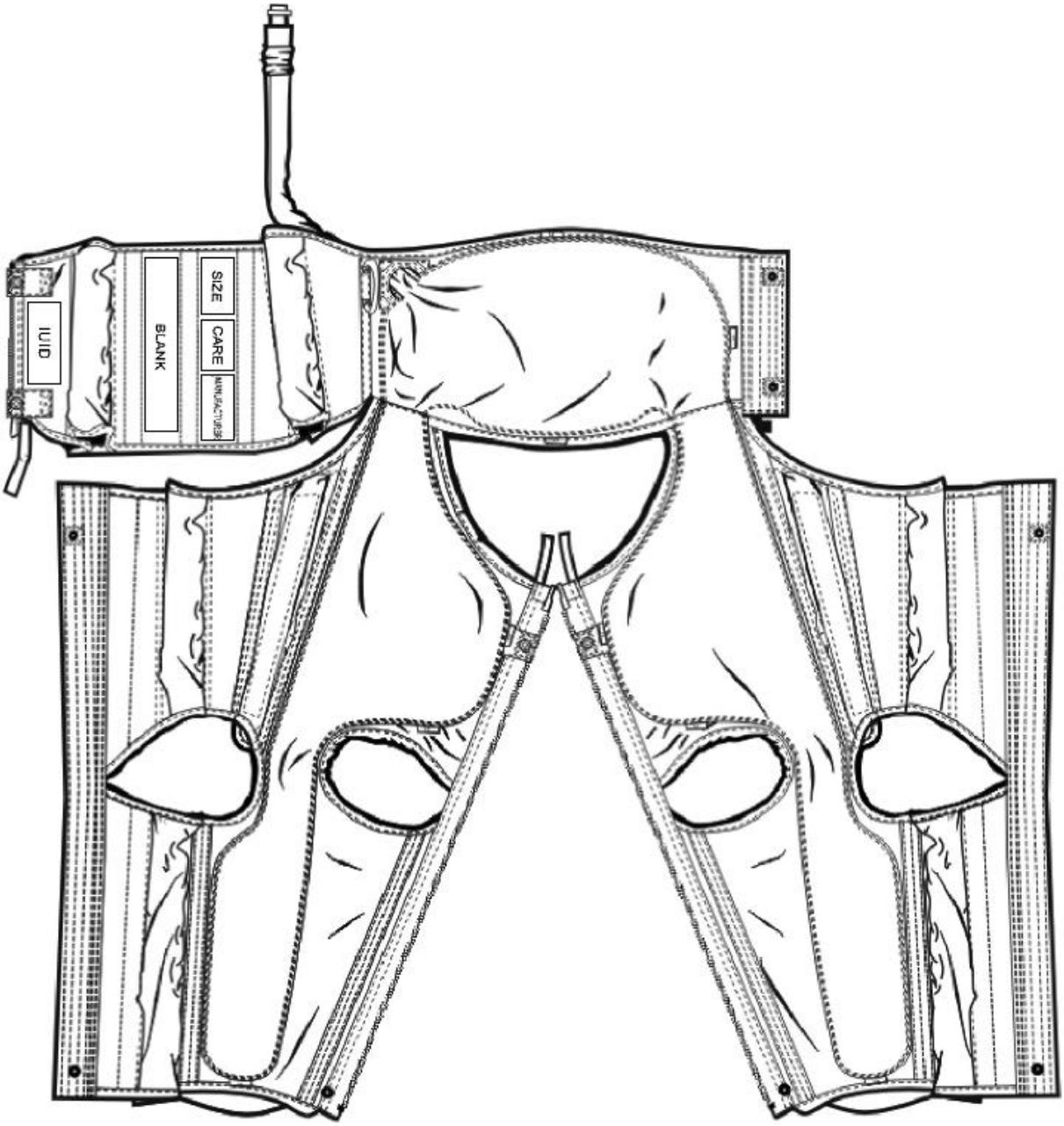


FIGURE 13A. Position of labels.

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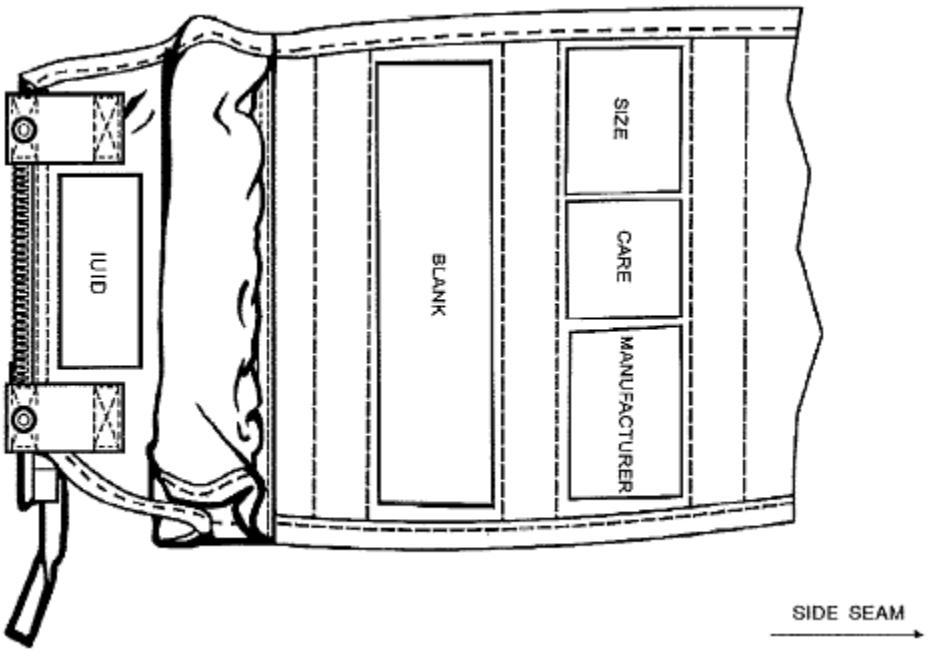


Figure 13B. Close-up label Position

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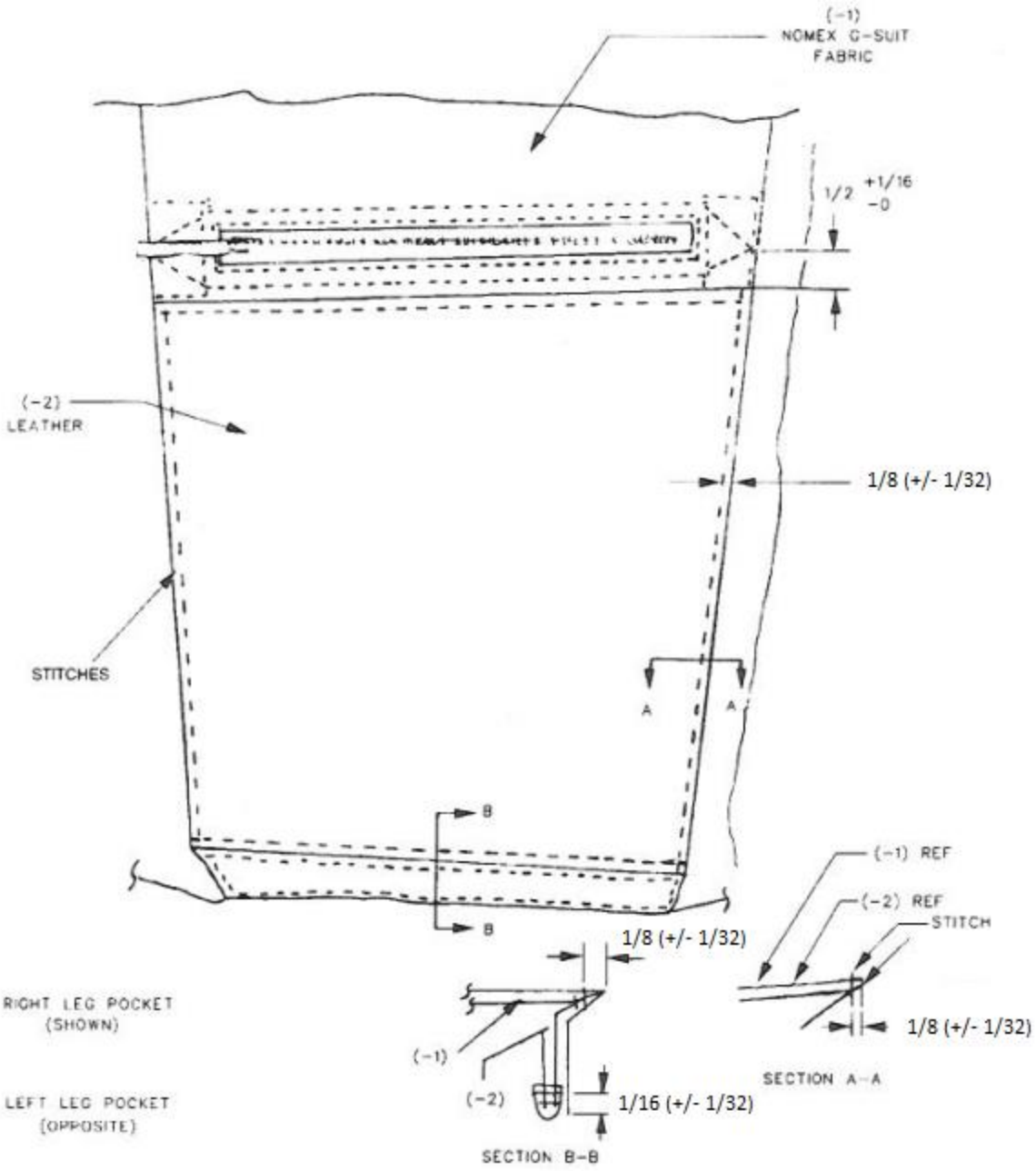


FIGURE 14. Leather calf pocket

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5. PACKAGING

5.1. Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.1.1. Special Markings

The shipping containers for the anti-g garments that have been subjected to the endurance test specified in 4.6.3 shall be marked with black or red waterproof ink in letters that are at least 1 inch high, with the following information: ENDURANCE TESTED – NOT TO BE USED IN FLIGHT.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1. Intended use.

The anti-g garment covered by this specification is intended to be worn by fighter aircraft pilots to counteract positive head to foot G forces. The extreme g-forces experienced by fighter aircraft pilots makes this a critical life support item as well as a military unique item.

6.2. Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of this specification, including any amendments.

b. Size required (see 1.2).

c. When first article is required (see 3.1).

d. Packaging requirements (see 5.1).

e. Government furnished patterns (see 3.11).

6.3. Qualification.

With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable Qualified Products Database (QPD), whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from AFLCMC/WNUS, Robins AFB, GA, 31098.

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6.4. Color samples.
Samples of color shades can be obtained from the procuring activity or as directed by the contracting officer. US Air Force shade Sage Green 1659 is equivalent to US Navy shade Green 3546. US Air Force shade Tan 1657 is equivalent to US Navy shade Khaki 3729.

6.5. Test samples.
The test methods described in 4.6.3 for endurance and in 4.6.8 for bond strength (bladder), which are required for the sampling inspection specified in 4.5.2.2 and 4.5.2.3, are destructive tests. Since these tests are destructive, the anti-g garments selected and tested in accordance with 4.5.2.2 and 4.5.2.3 should not be included in the quantities of serviceable finished end items required. The anti-g garments selected as samples and subjected to the endurance test and the bladders selected as samples and subjected to the bond strength (bladder) test needs to be delivered to the procuring activity, when requested. Any anti-g garment that has experienced more than 100 inflation test cycles at 11 psi or above needs to be deemed as having undergone destructive testing. This garment is not acceptable for shipment against the quantities of serviceable finished end items required by contract. Samples not shipped to the procuring activity needs to be retained by the contractor for the remainder of the contract.

6.6. Government-furnished property.
The contracting officer should arrange to furnish the patterns listed in 3.11.

6.7. Adhesive source.
UR-1092 Clifton Adhesive Part No. 7531037; 282 Burgess Place, Wayne, NJ 07474-0282

6.8. Shelf-life.
This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent of shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, Shelf-life Management Manual. The shelf-life codes are in the Federal Logistics Information System (FLIS) Total Item Record. Additive information for shelf-life management may be obtained from DoD 4120.27-M, or the designated shelf-life Points of Contact (POC). The PoC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate PoCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.hq.dla.mil/.

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6.9. Subject term (key word) listing.

Aircrew

Aramid

Bladder

Bond Strength

Critical Life Support Item

High Performance Aircraft

High Temperature Storage

Inflation Tube Assembly

Low Temperature Storage

6.10. Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:

Air Force - 11

Preparing Activity:

Air Force – 11

(Project 8475-2023-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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